

SUPPLEMENT TO “NEURAL BELLMAN OPERATORS”

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The new Section S.15 proves the all-response service enclosure, joint procurement selection, signed opportunity-exposure identities, and the witness-reconstruction arguments. This supplement gives complete proofs for Neural Bellman Operators and its economic applications. The new dynamic section proves the signed transport criterion on feasible reachable states, decomposes the adjustment option into current and future continuation gains, and establishes the joint law–benefit–fee–permission certificate. It distinguishes active counterfactual surrender from implementation with an inactive stop option and derives the minimum procurement cost when both compulsory terms and financial capacity are priced. The replication account records coefficient arrays, lower-policy identities, support sets, numerical error allowances, and executed adverse interventions. All earlier operator, approximation, preference, transition-law, regional certification, participation, permission, recursive-utility, equilibrium, and computational proofs remain included. Statements concerning stored-array evaluation are separate from transfer to a different constructor or diffusion target.

S.1. OPERATOR PROOFS AND QUANTITATIVE APPROXIMATION

S.1.1. *Exact Policy Improvement*

We use the assumptions of the main paper’s exact separated-operator proposition. In particular, $I(p)(t, s)$ is a selector maximizing $\mathcal{H}^a[E(p)](t, s)$, not a stationary point of a parameterized actor objective. Continuity of E is in the norm controlling the value, time derivative, first spatial derivatives, second spatial derivatives, and the stated boundary traces. The admissible compact policy class is closed under I . These are assumptions on an exact boundary-value problem, not universal properties of neural policy classes.

PROOF OF EXACT SEPARATED-OPERATOR CONSISTENCY: Write $V_n = E(p_n)$ and $p_{n+1} = I(p_n)$. The evaluation equation gives $V_{n,t} + \mathcal{H}^{p_n}[V_n] = 0$. Pointwise maximization implies

$$V_{n,t} + \mathcal{H}^{p_{n+1}}[V_n] \geq 0.$$

The assumed comparison and verification inequalities, with the same boundary operator for both policies, imply $V_{n+1} \geq V_n$. Precompactness of the evaluation image implies boundedness and hence a pointwise limit $\bar{V}$. Every subsequential limit in the declared strong topology has this same value component. Its derivative components are the derivatives of that value, because uniform convergence of the functions and their derivatives preserves differentiation. Consequently all strong accumulation points coincide and $V_n \rightarrow \bar{V}$ strongly.

Let $p_{n_j} \rightarrow p$ be a policy subsequence, which exists by compactness. Continuity of evaluation gives $E(p) = \bar{V}$. Continuity of improvement at p and continuity of evaluation give

$$E(I(p)) = \lim_j E(I(p_{n_j})) = \lim_j V_{n_j+1} = \bar{V}.$$

Evaluating $I(p)$ at this common value yields

$$0 = \bar{V}_t + \mathcal{H}^{I(p)}[\bar{V}] = \bar{V}_t + \sup_a \mathcal{H}^a[\bar{V}].$$

The boundary traces also pass to the limit. Comparison identifies $\bar{V}$ with the optimal value, and verification identifies p as optimal because $E(p) = \bar{V}$. This argument does not require $p = I(p)$ or convergence of the entire policy sequence. In the presence of tied maximizing actions, different optimal policies may share the same limiting value. Q.E.D.

The proof is an application of exact policy-improvement reasoning; it is not a claim of priority over [Jacka and Mijatović \(2015\)](#). The assumption of a compact strongly regular evaluation image can fail at corners or under degenerate diffusions. The stopped application is therefore evaluated by its specified positive transition approximation rather than asserted to satisfy every classical hypothesis globally.

S.1.2. Finite-Horizon Value-Loss Bound

For clarity, put $\Gamma_{n,n} = 1$ and $\Gamma_{n,j} = \prod_{\ell=n}^{j-1} \gamma_\ell$. Both $\mathcal{T}_n^a$ and its policy version are monotone and γ_n -Lipschitz in the sup norm. The implemented operator is $\tilde{\mathcal{T}}_n^a$. The three nonnegative error bounds at $\hat{v}_{n+1}$ are

$$\begin{aligned} \epsilon_n &\geq \left\| \hat{v}_n - \tilde{\mathcal{T}}_n^{\hat{p}_n} \hat{v}_{n+1} \right\|_\infty, \\ \delta_n &\geq \left\| \tilde{\mathcal{T}}_n \hat{v}_{n+1} - \tilde{\mathcal{T}}_n^{\hat{p}_n} \hat{v}_{n+1} \right\|_\infty, \\ \kappa_n &\geq \sup_a \left\| (\tilde{\mathcal{T}}_n^a - \mathcal{T}_n^a) \hat{v}_{n+1} \right\|_\infty. \end{aligned}$$

Here $\tilde{\mathcal{T}}_n = \sup_a \tilde{\mathcal{T}}_n^a$, and $\|\hat{v}_N - G\|_\infty \leq \eta$.

PROOF OF THE MAIN VALUE-LOSS THEOREM: Let $D_n = \left\| \hat{v}_n - J_n^{\hat{p}} \right\|_\infty$. Inserting the implemented and target fixed-policy backups gives

$$D_n \leq \epsilon_n + \kappa_n + \gamma_n D_{n+1}.$$

Let $R_n = \|\hat{v}_n - V_n^*\|_\infty$. The supremum inequality $|\sup_a f_a - \sup_a g_a| \leq \sup_a |f_a - g_a|$ and the nonnegative improvement gap give

$$R_n \leq \epsilon_n + \delta_n + \kappa_n + \gamma_n R_{n+1}.$$

Backward iteration, with $D_N, R_N \leq \eta$, proves

$$\begin{aligned} D_n &\leq \sum_{j=n}^{N-1} \Gamma_{n,j} (\epsilon_j + \kappa_j) + \Gamma_{n,N} \eta, \\ R_n &\leq \sum_{j=n}^{N-1} \Gamma_{n,j} (\epsilon_j + \delta_j + \kappa_j) + \Gamma_{n,N} \eta. \end{aligned}$$

Feasibility implies $V_n^* \geq J_n^{\hat{p}}$. Combining the two inequalities proves the stated upper bound with coefficients $2\epsilon_j + \delta_j + 2\kappa_j$ and terminal coefficient 2η . Q.E.D.

The theorem is finite-horizon. It permits a recursive operator with $\gamma_n > 1$ but then records amplification explicitly through $\Gamma_{n,j}$. It does not imply contraction for an arbitrary Epstein–Zin aggregator. Boundary killing can be included in the transition kernel by treating the boundary payoff as a known reward component; only continuation mass contributes to γ_n .

S.1.3. Differential Residual and Boundary Error

Let v be a $C^{1,2}$ candidate for the additive, stopped, discounted problem. Define $R^a = v_t + \ell^a + \mathcal{L}^a v - \rho v$. Suppose $|R^{\hat{p}}| \leq \epsilon(t)$, $\sup_a R^a - R^{\hat{p}} \leq \delta(t)$, and the payoff discrepancy at terminal time or exit is at most η . Assume the localized Itô formula passes to expectations by uniform integrability.

For every admissible p , stopped Itô integration gives

$$J^p(t, s) - v(t, s) = \mathbb{E}_{t,s}^p \left[\int_t^\tau e^{-\rho(r-t)} R^{p_r}(r, S_r) dr + e^{-\rho(\tau-t)} (G - v)(\tau, S_\tau) \right].$$

The upper bound for an arbitrary policy is $\eta + \int_t^T e^{-\rho(r-t)} (\epsilon(r) + \delta(r)) dr$. The absolute discrepancy for $\hat{p}$ is at most $\eta + \int_t^T e^{-\rho(r-t)} \epsilon(r) dr$. Taking a supremum over p and adding proves the differential corollary. The argument is conditional on uniform residual and boundary bounds. Replacing these bounds by sampled root mean squared losses is not a valid proof step.

S.1.4. Feasible Candidates, Convex Gaps, and Reference Bounds

If $Q(s, a)$ is L_a -Lipschitz in a and a feasible set $\mathcal{C}(s)$ is an h_a -net of the admissible actions, then a maximizer over $\mathcal{C}(s)$ is within $L_a h_a$ of the full supremum. Adding a neural proposal to $\mathcal{C}(s)$ cannot reduce its achieved backup value. Neither statement bounds the evaluation error of the critic that generated Q .

For differentiable convex F on a compact convex set A , its tangent inequality implies

$$F(a) - \inf_{b \in A} F(b) \leq \nabla F(a)^\top a - \min_{b \in A} \nabla F(a)^\top b =: g_F(a).$$

For $A = \{a \geq 0 : \sum_i a_i \leq B\}$, the minimum is $B \min(0, \min_i \partial_i F(a))$. The same formula applies to a scenario tree with separate feasible actions at each information node: sum the node-specific linear minimization gaps, using the gradient of the complete probability-weighted objective. This gradient must include all descendant costs.

If a feasible reference has objective C_R and gap G_R , then $C_R - G_R \leq C^* \leq C_R$. Any other feasible policy, evaluated on the same tree, therefore satisfies

$$C_\pi - C_R \leq C_\pi - C^* \leq C_\pi - C_R + G_R.$$

The program evaluates both objectives by all 64 terminal branches. It does not insert the neural critic as a substitute for policy cost. The reported interval is based on ordinary double-precision evaluation of these inequalities; it is not an interval-arithmetic enclosure of rounding error.

A useful conditional policy bound also follows. If the optimal continuation backup $Q^*(s, \cdot)$ is μ -strongly concave and the approximate continuation is within e of the optimal one, an action whose approximate improvement gap is at most δ obeys

$$\|\hat{a} - a^*\|^2 \leq \frac{2}{\mu}(\delta + 2\gamma e).$$

Indeed the value of each action changes by at most γe , and strong concavity bounds the resulting true suboptimality below by $\mu \|\hat{a} - a^*\|^2 / 2$. This is not applied to a nonconcave NDU maximand. In the convex resource problem the immediate quadratic cost supplies strong convexity $0.2/d$ for minimization. Its deterioration with dimension is included in the conditioning discussion rather than hidden in a dimension-free speed claim.

S.2. COUNTEREXAMPLES, TRACE ESTIMATION, AND DIFFERENTIATION

S.2.1. *Why a Joint Penalty Does Not Define the Economic Solution*

Consider $-V' + V = 0$ on $[0, 1]$, $V(1) = 1$. Its solution is $V^*(t) = e^{t-1}$. The historical composite objective was of the form

$$L(V) = \int_0^1 \{V(t) + [-V'(t) + V(t)]^2\} dt.$$

The feasible perturbation $V_\varepsilon(t) = V^*(t) - \varepsilon(1 - t)$ preserves the terminal condition but has residual $-\varepsilon(2 - t)$. Consequently

$$L(V_\varepsilon) - L(V^*) = -\frac{\varepsilon}{2} + \frac{7\varepsilon^2}{3} < 0 \quad \text{for } 0 < \varepsilon < \frac{3}{14}.$$

Thus the exact economic solution is not even a local minimizer of this penalty. Separate parameter blocks, objectives, and verification conditions are substantive changes, not a change of notation for that joint loss.

S.2.2. *A Stable Actor Stationary Point Need Not Be Optimal*

On $[-2, 2]$, let $H(a) = -(a^2 - 1)^2 + 0.2a$. Its stationary equation is $4a^3 - 4a - 0.2 = 0$. There are three real roots. At the negative local maximizer, approximately -0.9739943532 , $H''(a) = 4 - 12a^2 < 0$, so gradient ascent has a locally asymptotically stable equilibrium. The positive maximizer is approximately 1.0241203002 and has higher payoff by approximately 0.3998745872 . Both are isolated. The example embeds into a finite-horizon control problem with flow $H(a)$ and exact evaluation $(T - t)H(a)$.

Consequently, even exact fast evaluation, diminishing separated step sizes, bounded iterates, and zero noise do not turn isolated actor stationary points into policy-improvement fixed points. Under a suitable stochastic-approximation theorem the justified conclusion concerns the limiting invariant or stationary set. Global optimality additionally requires an argument excluding suboptimal attractors. Convex improvement in the resource example supplies such an argument for each surrogate action problem; it does not automatically eliminate critic approximation error. Constant-step Adam has no convergence theorem in this paper, and the executed neural runs use the explicitly recorded L-BFGS and constrained least-squares routines instead.

S.2.3. Small Strong Residuals Can Miss a Viscosity Solution

For the exit-time problem $dX_t = a_t dt$, $|a_t| \leq 1$, with flow -1 and zero payoff on $\{-1, 1\}$, the value is $V(x) = |x| - 1$. The viscosity equation is $1 - |V'| = 0$. The smooth functions

$$W_\varepsilon(x) = \sqrt{1 + \varepsilon^2} - \sqrt{x^2 + \varepsilon^2}$$

satisfy the boundary data, while their squared strong residual under the uniform distribution tends to zero:

$$\frac{1}{2} \int_{-1}^1 \left(1 - \frac{|x|}{\sqrt{x^2 + \varepsilon^2}}\right)^2 dx \sim \left(2 - \frac{\pi}{2}\right) \varepsilon.$$

Nevertheless $W_\varepsilon \rightarrow 1 - |x|$, which has the wrong sign and is not the desired viscosity solution. The asymptotic expression follows by $x = \varepsilon y$ and integration of the resulting integrable function on $[0, \infty)$. The code also evaluates the finite- ε integral by independent adaptive quadrature.

Monotonicity, stability, and consistency with an appropriate comparison principle are central to viscosity approximation; see [Barles and Souganidis \(1991\)](#). Smooth network approximation of a function and its derivatives, as in [Hornik, Stinchcombe, and White \(1990\)](#), concerns a smooth target and does not approximate a derivative at a kink or replace a viscosity-selection argument. Historical kink and residual arrays remain available as archival material, not as evidence that an unregularized strong-residual loss selects the correct solution.

S.2.4. Probe Budget and the Squared-Loss Estimand

Let $A = B^\top \nabla^2 v B$ be symmetric, z_k independent standard Gaussian vectors, and $q_k = z_k^\top A z_k$. Gaussian fourth moments give $\mathbb{E} q_k = \text{tr } A$ and $\text{Var}(q_k) = 2 \|A\|_F^2$. For deterministic a and $\bar{q}_K = K^{-1} \sum_k q_k$,

$$\mathbb{E} \left(a - \frac{1}{2} \bar{q}_K \right)^2 = \left(a - \frac{1}{2} \text{tr } A \right)^2 + \frac{\|A\|_F^2}{2K}.$$

In contrast $K^{-1} \sum_k (a - q_k/2)^2$ has excess expectation $\|A\|_F^2/2$, independent of K . For the regression matrix $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, $\|A\|_F^2 = 15$, $a = 1.25$, and the exact-trace loss is 1.5625. The population excess is $7.5/K$.

An unbiased alternative for $K \geq 2$ is the off-diagonal statistic

$$U_K = \frac{(\sum_k r_k)^2 - \sum_k r_k^2}{K(K-1)}, \quad r_k = a - q_k/2.$$

Independence implies $\mathbb{E} U_K = (\mathbb{E} r_k)^2$. Under an integrable derivative envelope, differentiating its expectation yields the gradient of the exact squared residual. The realized statistic can be negative; unbiasedness neither makes it a nonnegative sample loss nor guarantees optimizer stability. If the action or parameters are selected using the same probes, the fixed-parameter identity is not an adaptive generalization theorem. Independent held-out probes are needed to interpret such selection.

TABLE S.1
INDEPENDENT-BATCH STOCHASTIC-TRACE LOSSES

K	Correct mean	Population	MC s.e.	Old mean	Debiased mean
1	8.91810	9.06250	0.22234	8.91810	–
4	3.47199	3.43750	0.04758	9.17095	1.57233
16	2.03146	2.03125	0.01492	9.06287	1.56270
64	1.68023	1.67969	0.00640	9.06132	1.56307

20,000 independent Gaussian batches per row; the exact-trace squared residual is 1.5625. Correct mean averages probes before squaring; old mean averages individual squared residuals. The debiased off-diagonal statistic requires at least two probes. Its standard errors and both variance estimands are retained in the JSON record.

Each row averages 20,000 independent batches. The Monte Carlo standard error pertains to the repeated-batch mean, not to a single quadratic form or its within-batch variance. The numerical test checks the known expectation within six Monte Carlo standard errors and preserves the wrong mean-of-squares statistic as a negative control. It does not assign value or policy errors to this algebraic experiment.

S.2.5. Executable Computational-Graph Tests

The test suite forms a quadratic critic and a control-dependent diffusion loading, contracts its automatic Hessian with the covariance, and checks both parameter blocks. In the actor update, critic coefficients receive no gradient but the action-dependent covariance does. In the critic update, the policy output is detached while the critic coefficients receive gradients. A finite-transition actor derivative is also checked against central differences in all three controls. Freezing critic parameters does not detach the next-state input: the continuation gradient with respect to the action-induced state is retained.

A separate test compares the resource critic’s explicit gradient with PyTorch automatic differentiation. The convex squared-ReLU feature architecture is a transparent special case of the input-convex approach of [Amos, Xu, and Kolter \(2017\)](#). Exact directional second derivatives or Hessian-vector products, following [Pearlmutter \(1994\)](#), can avoid constructing a dense Hessian when the diffusion has low rank. The complete costs still depend on network size, diffusion multiplication, sample count, probe count, and optimization iterations.

S.3. THE STOPPED PREFERENCE-FORMATION ECONOMY

S.3.1. Admissibility and the Boundary Contract

The interior state is $(u, X) \in (1.2, 2.8) \times (0.5, 2)$, with the controls and coefficients stated in the main paper. The process is stopped at the first exit or $T = 1$, whichever occurs first. It receives $G(u, X) = -0.02(u - 2)^2 + 0.1 \log X$ at that time. A start on the boundary is settled immediately. This is a finite-horizon contract on an open operating domain; it is not a claim that the unstopped diffusion remains in a rectangle.

For bounded progressively measurable controls, the wealth SDE with coefficients linear in wealth and the additive preference SDE have a well-defined solution up to exit. Before exit, wealth and consumption are bounded away from zero. The flow, terminal payoff, and quadratic adjustment cost are bounded on the stopped domain. Thus the

additive objective is finite without introducing a reflected local time or a wealth-transfer process.

The original state-constraint interpretation cannot hold with these coefficients. Preference volatility is 0.05 at both endpoints, and the maximal wealth drift at the lower boundary is $(0.02 + 0.8 \times 0.06)0.5 - 0.05 = -0.016$. At zero portfolio volatility it is more negative. Neither fact is repaired by projecting an Euler endpoint back onto the rectangle. The stopped contract retains all interior coefficients and makes the change in boundary economics explicit.

S.3.2. A Boundary-Compatible Payoff

The choice of G has a useful analytic check. Its derivatives are $G_u = -0.04(u - 2)$, $G_{uu} = -0.04$, $G_X = 0.1/X$, $G_{XX} = -0.1/X^2$, and $G_{uX} = 0$. For every admissible action and every $k \geq 0$,

$$\begin{aligned} \ell^a + \mathcal{L}^a G - \rho G &\leq \max_{u \in [1.2, 2.8]} \frac{0.8^{1-u}}{1-u} + 0.0064 + 0.0043 - 0.00005 + 0.0032846 \\ &< -0.8162. \end{aligned}$$

The consumption bound follows because utility is increasing in consumption. Maximization over u at $c = 0.8$ gives the upper endpoint; it can also be checked by the explicit derivative. The 0.0043 term bounds $0.1[0.02 + 0.06\pi - c/X]$ by using $\pi \leq 0.8$, $c/X \geq 0.025$. The omitted portfolio diffusion term is nonpositive, and so are adjustment costs. The last term bounds $-\rho G$ uniformly.

Stopped Itô integration therefore implies $J^p \leq G$ for every admissible policy. To obtain boundary continuity from below, fix for example a constant admissible nonzero portfolio share. Its two-dimensional covariance is uniformly positive definite on the closed rectangle because $X \geq 0.5$ and $|\varrho| < 1$. The boundary is regular for this fixed diffusion, and its expected exit time tends to zero as the start tends to a boundary point. Bounded rewards and continuity of G imply that this policy's stopped payoff tends to G there. Since its payoff is bounded above by the optimal value and the latter is bounded above by G , the value has the declared boundary limit. This argument establishes boundary compatibility; it does not assert global classical differentiability at corners.

S.3.3. Transition, Covariance, and First Exit

For step h , form four equiprobable sign pairs $(z_1, z_2) \in \{-1, 1\}^2$. Let

$$\begin{aligned} d_u &= \theta h + \sigma_u \sqrt{h} z_1, \\ d_X &= [(r + \pi(\mu - r))X - c]h + \pi X \sigma_X \sqrt{h} [\varrho z_1 + \sqrt{1 - \varrho^2} z_2]. \end{aligned}$$

For an interior state s , α is the first fraction in $[0, 1]$ at which the line $s + \alpha d$ meets a face, or one if it stays inside. An exit branch has payoff

$$\frac{1 - e^{-\rho \alpha h}}{\rho} \ell(s, \alpha) + e^{-\rho \alpha h} G(s + \alpha d).$$

A surviving branch instead uses the continuation value at $s + d$ and the full-step flow annuity. At a boundary start the backup is exactly $G(s)$. In particular there is no nonzero continuation from a boundary node. The reference evaluator uses nonnegative bilinear interpolation weights that sum to one on surviving branches.

These signs reproduce conditional first moments and the covariance matrix of the written diffusion. Polynomial tests on $1, u, X, u^2, X^2, uX$ distinguish covariance terms from drift-product terms of order h . At the test state and action the covariance is

$$\begin{pmatrix} 0.0025 & -0.002 \\ -0.002 & 0.0256 \end{pmatrix}.$$

The largest generator discrepancy decreases by a factor of ten when h decreases from 10^{-2} to 10^{-3} and again to 10^{-4} . Preference controls of -0.2 and 0.2 give different mean next states even when both displacements are smaller than a spatial grid cell. No nearest-neighbor map erases the control.

The line-exit rule is an approximation to diffusion killing, not exact Brownian bridge detection. Before-correction overshoots, branch-exit frequencies, occupied exit probabilities, and post-exit payoffs are distinct recorded objects. A reflection-regulator field is inapplicable and is represented by null with that explanation. For the joint state–time refinement, the squared largest mesh width divided by h decreases as $0.04, 0.02, 0.01$; the separate control refinements are also retained. These observations support consistency diagnostics but do not supply a numerical bound for every κ_n in the main theorem.

S.3.4. Economic Comparative Statics

For each admissible policy let $B(p)$ denote cost-free benefit. Define discounted cumulative effort by $C(p) = \mathbb{E} \int_0^{\tau_p} e^{-\rho t} \theta_t^2 / 2 dt$. The stopping time is policy dependent but not directly dependent on k for a fixed policy. Thus $V(k) = \sup_p [B(p) - kC(p)]$ is decreasing and convex. For optimizers p_1, p_2 at $k_1 < k_2$, revealed optimality gives

$$\begin{aligned} B(p_1) - k_1 C(p_1) &\geq B(p_2) - k_1 C(p_2), \\ B(p_2) - k_2 C(p_2) &\geq B(p_1) - k_2 C(p_1). \end{aligned}$$

Adding implies $(k_2 - k_1)[C(p_2) - C(p_1)] \leq 0$. Also $-C(p)$ is a subgradient of V at any k for which p is optimal, and $V'(k) = -C(p)$ wherever V is differentiable. For policies with value losses at most e_1, e_2 , the same inequalities with these errors imply

$$C(p_2) - C(p_1) \leq \frac{e_1 + e_2}{k_2 - k_1}.$$

The result applies to the continuous stopped model and separately to each exactly specified positive finite approximation. A finite-grid comparison cannot set $e_i = 0$ for the continuous problem merely because its discrete maximization was exhaustive.

The reported effort is computed both backward and by forwarding the induced state probabilities. The latter multiplies live branch mass by bilinear weights and accumulates discounted cost through the line-exit fraction. This is an occupation calculation for the interpolation-defined finite Markov chain. It provides an economically interpretable sum rather than an unweighted mean of control magnitudes. A reduction of this aggregate does not imply a pointwise reduction of θ , consumption, or portfolio share. Nor does a comparative static identify k from observations without a specified data model and an injective moment map.

S.4. PORTFOLIO AND RECURSIVE-UTILITY VERIFICATION

S.4.1. *Stationary Merton Benchmark*

Write $b = 1 - \gamma$, $V(X) = AX^b/b$, $c = mX$, and

$$g(m, \pi) = r + (\mu - r)\pi - m + \frac{1}{2}(b - 1)\sigma^2\pi^2.$$

For exponential utility discount ρ , the fixed-policy stationary equation is

$$A = \frac{m^b}{\rho - bg(m, \pi)}$$

on the positive branch $\rho - bg > 0$. Optimization gives $\pi^* = (\mu - r)/(\gamma\sigma^2)$ and

$$m^* = \frac{\rho - (1 - \gamma)[r + (\mu - r)^2/(2\gamma\sigma^2)]}{\gamma}.$$

At $(\rho, \gamma, \mu, r, \sigma) = (0.04, 2, 0.08, 0.02, 0.2)$ the targets are $(m^*, \pi^*) = (0.04125, 0.75)$. Wealth remains positive under this proportional policy. Its discounted terminal candidate expectation decays at rate $\rho - bg(m^*, \pi^*) = m^* > 0$, verifying its own transversality condition. The usual verification comparison is made over policies for which the stopped local martingale and discounted terminal inequalities pass to the limit.

The executed feature solver treats $q = \log A$, the consumption ratio, and the portfolio ratio as unknowns. It alternates the fixed-policy scalar evaluation root with an automatic-differentiation actor optimization, without feeding the analytical targets into the updates. Positivity is imposed by $A = e^q$. This is a learned homothetic feature model; the ratios alone would not be a neural-function benchmark on an unrestricted state space. The no-short regression at $\mu = 0.01$ instead has unconstrained share -0.125 and constrained share zero at every positive wealth. It is not a wealth-dependent switching problem.

S.4.2. *Epstein–Zin Driver and the Admissible Comparison Class*

Use $a = 1 - 1/\psi$, $b = 1 - \gamma$, with $a, b \neq 0$, $c > 0$ and $bV > 0$. The driver is

$$f(c, V) = \frac{\rho}{a} \{c^a (bV)^{1-a/b} - bV\}.$$

The log limits must be taken separately when a denominator vanishes. For a homothetic candidate $V = AX^b/b$, dividing the stationary HJB by AX^b gives

$$\frac{\rho}{a} \{m^a A^{-a/b} - 1\} + g(m, \pi) = 0.$$

Thus a fixed proportional policy has a unique positive homothetic coefficient whenever $1 - ag/\rho > 0$. Its consumption first-order condition is $\rho m^{a-1} A^{-a/b} = 1$, and the portfolio first-order condition gives the same Merton share with risk aversion γ . Eliminating A yields

$$\pi^* = \frac{\mu - r}{\gamma\sigma^2},$$

$$m^* = \psi\rho + (1 - \psi) \left[r + \frac{(\mu - r)^2}{2\gamma\sigma^2} \right],$$

$$A^* = (\rho(m^*)^{a-1})^{b/a}.$$

For $(\rho, \gamma, \psi, \mu, r, \sigma) = (0.04, 5, 1.5, 0.08, 0.02, 0.2)$ this gives $m^* = 0.0455$ and $\pi^* = 0.3$. The normalized Hamiltonian is strictly concave in the consumption ratio and portfolio share on the specified positive domain. The selected ratios lie inside the implemented action bounds.

For precision about infinite horizon, the verification class consists of policies for which the policy-wise recursive equation exists on the branch $bY > 0$, its localized comparison argument is integrable, and the comparison-weighted terminal remainder vanishes. On compact subsets of $c > 0, bY > 0$ the driver is locally Lipschitz, but this does not imply a global Lipschitz constant on the entire half-line. One may first stop at compact localization sets, apply the mean-value linearization in the value argument, and then impose uniform integrability and the weighted terminal limit to remove localization. With $\Gamma_t = \exp(\int_0^t f_V(c_s, \zeta_s) ds)$ for the intermediate value ζ_s , the required remainder is of the form $\mathbb{E}[\Gamma_T |V(X_T) - Y_T^p|] \rightarrow 0$. The candidate HJB inequality and this comparison yield $Y_0^p \leq V(X_0)$, with equality for the maximizing policy.

For the selected homothetic policy itself, $f_V = -0.1115$, while the growth exponent of $\mathbb{E}X_t^b$ is $bg = 0.066$. The candidate comparison-weighted value therefore decays at rate $0.1115 - 0.066 = 0.0455 > 0$. The numerical record evaluates the fixed-policy equation and target errors, and the analytic calculation verifies this decay. This specifies a genuine stationary recursive benchmark with an explicit admissible verification class; it does not assert that all bounded controls satisfy the terminal condition automatically.

The sign-enforcing architecture is actually executed: $V = e^q X^b / b$, hence $bV = e^q X^b > 0$. A separate finite-horizon architecture test uses

$$V_\xi(t, X) = \frac{1}{b} \exp\{\log[bG(X)] + (T - t)N_\xi(t, X)\}$$

for $bG > 0$. It verifies the terminal identity and sign on test points. It is a domain test, not evidence that an additional finite-horizon recursive problem was trained. These distinctions follow the policy-wise recursive-utility formulation of [Duffie and Epstein \(1992\)](#) rather than replacing recursion by a pointwise evaluation at $c = 1, V = -1$.

S.4.3. A Structured Stochastic Dynamic Control Check

For $d \in \{4, 8, 16\}$, the additional six-period LQR problem uses

$$A = 0.82I + 0.07(J + J^\top), \quad B = 0.2I, \quad Q = I + 0.31\mathbf{1}^\top/d, \quad R = 0.15I,$$

and Gaussian shock loading $\Sigma = 0.08(I + 0.31\mathbf{1}^\top/d)$, with terminal matrix Q . Here J has ones on its upper adjacent diagonal. The linear neural actor and quadratic feature critic are trained by policy evaluation and improvement. The independent reference uses the finite-horizon Riccati recursion, including the noise contribution to the value constant. A second recursion evaluates the learned fixed policy rather than using its fitted critic as a performance score.

The experiment has cross-state dynamics and non-diagonal stochastic covariance but no binding control constraints. It is retained as a structured dynamic regression test. The main shared-budget resource experiment supplies the separately executed constrained case; neither is relabeled as an adaptive sparse-grid comparison.

TABLE S.2
STRUCTURED STOCHASTIC LQR REGRESSION

d	Actor coeff. error	Cost excess max.	Neural sec.	Riccati sec.
4	4.25e-06	3.03e-11	0.0571	0.00037
8	2.25e-06	2.14e-11	0.0724	0.00028
16	4.30e-06	3.84e-11	0.1330	0.00033

Six periods, linear actors and quadratic feature critics, independent Riccati reference, and separate frozen-policy evaluation. Cost excess is normalized per dimension in the implementation. This structured unconstrained regression is not the main binding-resource experiment and supplies no generic neural speed advantage.

S.5. TEMPORAL SELVES AND DYNAMIC COMPETITION

S.5.1. *Sophisticated Consumption–Savings*

Let one period have length Δ , $\delta = e^{-\rho\Delta}$, and terminal payoff $G(w) = \log w$. The acting self chooses $c \in (0, w)$ to maximize $\Delta \log c + \beta \delta V_{n+1}(w - c)$, whereas ordinary continuation evaluates the chosen policy as

$$V_n(w) = \Delta \log c_n(w) + \delta V_{n+1}(w - c_n(w)).$$

If $V_{n+1} = A_{n+1} \log w + B_{n+1}$, strict concavity gives

$$\begin{aligned} z_n &:= \frac{c_n(w)}{w} = \frac{\Delta}{\Delta + \beta \delta A_{n+1}}, \\ A_n &= \Delta + \delta A_{n+1}, \\ B_n &= \Delta \log z_n + \delta A_{n+1} \log(1 - z_n) + \delta B_{n+1}. \end{aligned}$$

Starting with $A_N = 1, B_N = 0$ yields an exact sophisticated equilibrium. The coefficient A_n is independent of β , but the policy and the constant are not. The reference test uses $\Delta = 1$, four consumption dates, $\rho = 0.04$, and $\beta \in \{0.7, 1\}$. At four wealth levels it separately solves the bounded one-shot consumption problem and checks the evaluation identity with the frozen continuation coefficients.

For two consumption dates, $\delta = 1$, the correct first-date ratio is $1/(1 + 2\beta)$. Reusing the acting self's maximand as its stored continuation would instead yield $1/(1 + \beta + \beta^2)$. At $\beta = 0.7$ the difference is 0.4166667 versus 0.4566210. The $\beta = 1$ check alone cannot reveal this error. These checks implement the written temporal game rather than an ordinary geometric-discount control problem. They do not identify the finite-state reference as a trained neural solution.

S.5.2. *A Stochastic Capacity Duopoly*

There are four decision dates. Demand is $D \in \{1, 1.3\}$ with probability 0.8 of staying at its current value. Each firm's installed capacity $K_i \in \{0, 1\}$ determines current output $q_i = K_i/3$. Current operating profit is $q_i(D - q_i - q_j)$. Firm i chooses a maintenance or renewal intensity $a_i \in [0, 1]$ at cost $0.4a_i^2/2$. Independently across firms conditional on actions, next-period capacity is one with probability a_i and zero otherwise. The discount factor is 0.95 and terminal value is $0.05K_i$. There are eight Markov states and four dates, so complete state enumeration is possible.

For fixed rival intensity, the continuation value is affine in a firm's own intensity. Its best response is therefore the projection onto $[0, 1]$ of its discounted marginal continuation gain divided by 0.4. The joint map is affine before projection, with slope determined by the cross difference of the four next-capacity continuation values. The program checks the product of the two own best-response slopes, iterates from multiple initial profiles, and compares their limits. This finite example has a contraction bound below one. It does not establish uniqueness in arbitrary dynamic games.

An independent backward best-response recursion freezes the rival's entire state-contingent policy at every subsequent date and allows the tested firm to deviate at all dates. Its difference from the profile value is dynamic exploitability. This is stronger than checking a single current Hamiltonian against an inconsistent continuation. The report records the maximum over all dates, states, and players, with a numerical tolerance of 10^{-10} . A separate automatic-differentiation test confirms that a player's loss does not differentiate the rival's parameters.

The static special case $q_i(1 - q_i - q_j)$ remains a useful negative control. Symmetric Nash output is $1/3$, whereas joint-profit output is $1/4$. At the latter profile, changing one firm's quantity to its own best response is profitable. A symmetric or jointly optimal actor graph is therefore not a substitute for a Nash calculation. The dynamic capacity model is a finite-state equilibrium computation; no competitive limit or general neural-game convergence is inferred from it.

S.5.3. Market-Size Normalization and a Conditional Small-Player Bound

To retain the economically relevant many-firm extension without conflating different economies, consider per-firm normalized output y_i and actual output $q_i = y_i/N$, with inverse demand $P = a - b\bar{y}$, $\bar{y} = N^{-1} \sum_i y_i$. Multiplying each firm's monetary payoff by N gives normalized flow $y_i P - C(y_i)$ under consistently scaled costs. Its direct strategic price derivative is $-by_i/N$. Investment, depreciation, and state transitions must be scaled and retained as well; this current-flow derivative by itself does not establish convergence of dynamic equilibria.

A precise conditional implication is available. Suppose that, against a fixed rival profile, the finite- N player's transition operators and a price-taking comparison operator are monotone and γ_n -Lipschitz, have the same terminal payoff, and differ uniformly by at most η_n/N on all policies and continuation values used in the comparison. Then a policy optimal for the price-taking problem loses at most

$$\frac{2}{N} \sum_{j=n}^{N-1} \Gamma_{n,j} \eta_j$$

relative to the finite- N best response. To prove it, backward iteration bounds the difference of the two optimal values by $N^{-1} \sum_j \Gamma_{n,j} \eta_j$, and bounds the two evaluations of that fixed policy by the same amount. Their sum gives the result. Uniform Lipschitz control of aggregate-state effects is one route to the operator hypothesis, but it must be checked in a specified model. This proposition concerns a unilateral deviation bound, not selection or convergence of a sequence of equilibria. It is theoretical extension material and is not recorded as an executed large- N experiment.

S.6. REPLICATION AND ADDITIONAL RESULTS

This section retains the foundational neural experiments and their full checkpoint provenance. The fifth-revision additions and independent rechecks are described in S.7–S.8.

S.6.1. *Source and Evidence Identities*

The authoritative R4 source record consists of `ECTA_R4.tex`, `SUPP_R4.tex`, their explicitly included sections, and `replication/r4/`. The base review is pinned at commit `79a7d84`. The replication manifest records the reachable source commit, full SHA-256 digests of the source files, package versions, and output identities. Source files are committed before their authoritative execution. A later result commit can add generated outputs without changing that source anchor.

Execution status, evidence class, and numerical-tolerance status are separate fields. A completed program is not automatically a solved control problem. The coupled resource runs have a specified held-out cost-loss target and independent reference gaps. The NDU comparison reports discrepancies and does not assign a passed uniform PDE certificate. Algebraic checks, domain tests, finite-state equilibria, and structured feature solvers each have their own evidence class. An unavailable continuous-time error bound or reflection regulator is null with an explanation, not a numerical zero.

Scientific digests exclude elapsed time and hardware measurements. File digests still identify complete files, including timing records. Floating-point algorithms, package versions, and BLAS implementations can change last digits across platforms. The source and saved weights permit direct policy replay independently of retraining. Timings include neural fitting and full evaluation for the stated initial-state batch; they do not imply a universal hardware comparison or unmeasured amortization advantage.

S.6.2. *Architectures and Optimization*

For the stopped preference model the safeguarded neural run uses separate two-hidden-layer tanh actors of width 32 and critics of width 64 at each of eight decision dates. The actor has three bounded output coordinates. The critic writes the value as G plus remaining time times a product vanishing on all spatial faces times its network output. The training sample contains 2,048 interior states. L-BFGS caps are 160 actor and 1,000 critic iterations with strong-Wolfe line search; actual function evaluations and iteration counts are recorded when produced by the optimizer. The implementation never trains on the independent grid reference’s values or policies.

The unsafeguarded pilot uses width-32 critics and 1,024 training states with a smaller critic budget. It is retained as a development baseline, not used as a matched causal ablation. The resource experiment supplies a matched ablation: the same fitted actor weights are evaluated with and without pointwise improvement at deployment. It holds training, initial states, shocks, and model primitives fixed.

For resources the convex critic is

$$q(x) + \ell^\top(x - \mathbf{1}) + c_0 + \sum_{j=1}^{6d+102} \alpha_j [\max\{w_j^\top(x - \mathbf{1}) + b_j, 0\}]^2, \quad \alpha_j \geq 0.$$

Directions consist of signed coordinate axes, signed aggregate directions, and 32 seeded Gaussian directions, each at offsets $-0.5, 0, 0.5$. The coefficients are fitted by bounded

TABLE S.3
ALL RESOURCE RUNS AND THE MATCHED DEPLOYMENT ABLATION

d	Seed	NBO upper loss	Proposal only	Reference gap	Binding at 0
4	101	6.45e-04	1.19e-02	8.32e-08	0.812
4	202	4.88e-04	8.14e-03	7.40e-08	0.938
4	303	5.23e-04	6.31e-03	7.02e-08	0.938
8	101	4.29e-04	9.54e-02	9.79e-08	0.938
8	202	4.61e-04	3.28e-02	9.97e-08	1.000
8	303	2.03e-04	3.89e-02	9.75e-08	1.000
16	101	5.41e-04	6.73e-02	9.54e-08	1.000
16	202	5.67e-04	6.81e-02	9.63e-08	1.000
16	303	5.24e-04	6.65e-02	6.02e-08	1.000

Each row uses the stored 32 initial states and every one of the 64 terminal shock branches. The reference gap is the complete scenario-tree linear minimization gap at the tighter audit tolerance. Binding is the fraction of initial actions exhausting the common budget. All declared dimension–seed combinations are included.

least squares with a 10^{-5} ridge term. The actor has one width-32 tanh layer and a slack-softmax output enforcing the common budget. Its 160-iteration L-BFGS fit distills feasible greedy actions. The convex pointwise improvement uses an explicit gradient-Lipschitz upper bound and checks its linear minimization gap rather than equating a stationary point with a maximum. All three seeds, 101, 202, and 303, are included at each dimension.

The independent scenario tree contains one root action, four next-date actions, and sixteen last-date actions for each tested initial state. Its objective sums all state and control costs and the 64 terminal costs with their probabilities. An independent backward adjoint computes the gradient. Reference optimization stops at a common-accuracy gap of 10^{-3} for timing and is separately continued to 10^{-7} for policy-cost auditing. Nonanticipativity is enforced by one action per information node, not one freely chosen action per complete shock path.

S.6.3. *Reproduction Order*

The repository README gives executable commands for the following sequence: generator, temporal, trace, utility, and game tests; stochastic grid refinements and re-solved adjustment-cost panels; the retained unsafeguarded neural pilot; all three safeguarded NDU runs; nine resource runs; independent frozen-policy and occupation analysis; computational-graph and checkpoint-replay tests; table generation; source/output manifest generation; and LaTeX builds. The table generator reads the deposited JSON records, so manually entered historical arrays cannot silently become current numerical evidence.

The source snapshot also retains the previous manuscripts, supplements, reports, reviewer diagnostics, and arrays unchanged. The preservation map locates their substantive topics in the new manuscript or this supplement. Retention is not endorsement of earlier incorrect claims. The development log records the standard-library module-name collision discovered during implementation, the corrected execution, and the budget difference of the pilot. Failed commands are not relabeled as successful scientific experiments.

S.6.4. *What Each Numerical Error Means*

For NDU, a reference value difference compares two specified finite approximations or evaluates a neural policy against a specified finite reference. The actor discrepancy reports consumption, preference adjustment, and portfolio separately because they have different units. A feasible Bellman gain is measured in continuation-value units and uses an independently evaluated frozen continuation. It is a lower bound on the maximum profitable one-step deviation over a richer action set, not an upper bound on all deviations.

For resources, the full stochastic tree gives the evaluated policy cost. The reference tangent gap bounds its unknown optimum on that same tree, resulting in the interval in S.1. The held-out initial states are explicitly stored, so the bound is not extrapolated to all states. The Merton and Epstein–Zin policy errors compare learned homothetic coefficients with analytical solutions, while LQR errors compare structured learned policies with an independent Riccati solution. Temporal and game exploitability use separate best-response calculations against the actual continuation being evaluated. Trace standard errors concern a repeated-batch scalar loss and are not policy errors.

S.7. STOPPED CONTRACTS AND VALUE-ENCLOSED POLICY REUSE

S.7.1. *Discounted Duration and Terminal Exposure*

For any realized stopping time $\tau \leq T$, direct integration gives

$$\rho \int_t^\tau e^{-\rho(v-t)} dv + e^{-\rho(\tau-t)} = 1.$$

This equality holds before conditioning and before optimizing. In the positive finite model, each branch satisfies the same identity with its actual live duration αh . Thus the independently evaluated exposure recursion is

$$Z_n^p(s) = z_n(s, p_n(s)) + K_n^{p_n} Z_{n+1}^p(s), \quad Z_N^p = 1,$$

where z_n is the discounted probability weight of settlement during the current branch and K_n^p contains only live discounted interpolation weights. Duration satisfies $A_n^p = a_n^p + K_n^p A_{n+1}^p$, $A_N^p = 0$. The row identity $\rho a_n^p + z_n^p + K_n^p \mathbf{1} = 1$ proves $\rho A_n^p + Z_n^p = 1$ by induction. This includes immediate settlement for an initial boundary node.

The program does not define Z as $1 - \rho A$ and then test the resulting tautology. It evaluates Z with its own killed-payment recursion and checks the identity for randomly selected contracts and stored-policy switches. It then compares the original four-feature payoff $B + mA - kC + \ell Z$ with the reduced payoff $\ell + B + (m - \rho\ell)A - kC$. The largest discrepancy and the source hash are recorded. For state paths generated by an optimal-policy selection, all changes $(m, \ell) \mapsto (m + \rho h, \ell + h)$ preserve the set of feasible optimal laws; this is the precise sense in which choices alone do not identify the two levels separately.

The monotonicity result concerns a common transition and feasible-policy class. Changing the shock correlation, the control bounds, or the stopping mechanism does not satisfy that condition. Nor does a different parameter-specific action menu necessarily preserve convexity of the optimized finite value in the original parameters. We therefore freeze the complete union menu over the contract interval before building the library.

S.7.2. Certified Anchors and Approximate Policy Features

Let a feasible anchor policy p_i have certified loss $c_n^i(s)$ at its anchor q_i . Exact evaluation gives the valid upper bound $\bar{V}_n^i = J_n^i(q_i) + c_n^i$. If evaluation instead supplies an interval $[J_n^i, \bar{J}_n^i]$, the valid anchor upper bound is $\bar{J}_n^i(q_i) + c_n^i$. The empirical maximum of sampled actor residuals cannot be substituted for either ingredient.

For clarity, an error version of switching can be stated without conflating approximate policy features with exact evaluation. Suppose $\|\hat{J}_n^i - J_n^i\|_\infty \leq e_n$ uniformly over stored policies at the query, and choose the policy index maximizing $\hat{J}_n^i$. Let $L_n = \max_i J_n^i$ and $D_n = \|(L_n - J_n^{\text{sw}})_+\|_\infty$. If continuation kernels have row sums at most γ_n , then

$$D_n \leq 2e_n + \gamma_n D_{n+1}, \quad D_n \leq 2 \sum_{j=n}^{N-1} \Gamma_{n,j} e_j \quad (\text{S.1})$$

with common exact terminal evaluation. To prove the first inequality, the selected policy has exact value at least $L_n - 2e_n$. The next switching value is at least the selected continuation value minus D_{n+1} ; positivity gives the result. Hence

$$V_n^* - J_n^{\text{sw}} \leq U_n - L_n + 2 \sum_{j=n}^{N-1} \Gamma_{n,j} e_j.$$

The exact-feature theorem in the main paper is the $e_n = 0$ case. The executed finite recursion uses this case up to recorded floating-point errors; it does not assign zero error to the neural critic. This distinction allows neural approximation to provide proposals while the certificate uses separately evaluated policies.

S.7.3. A Finite Certificate for an Entire Interval

For adjacent anchors $a < b$, two feasible endpoint-policy lines are $b_1 + da_1$ and $b_2 + da_2$, independently at every date and state. With an affine upper chord $u_0 + du_1$, define

$$g(d) = u_0 + du_1 - \max\{b_1 + da_1, b_2 + da_2\}.$$

On either side of the intersection of the two lines, g is affine. Its maximum on $[a, b]$ must therefore occur at a , at b , or at their intersection if present. Parallel lines require only the endpoints. Taking the maximum over the finite set of dates and states gives a uniform bound without discretizing the contract interval. Enlarging the policy library can only improve the maximum of feasible policy values; using just the endpoint pair to certify an interval is conservative and reduces certification cost.

The adaptive construction begins with contracts zero and one. At every iteration it computes the complete bound, selects a worst interval, and solves a new anchor at a parameter attaining that interval's largest gap. Each new policy is independently evaluated. The complete anchor sequence, worst gaps, and parameter locations are saved, including the stopping criterion. The implementation has a maximum-anchor safety limit that raises an error rather than publishing an uncertified library when the tolerance is not reached.

All computations use double precision. The pairwise-intersection calculation treats duration-slope differences below 10^{-14} as parallel. The omitted variation of a difference of such lines on the unit contract interval is at most 10^{-14} ; the verification checks

use a larger 10^{-9} slack. The published 10^{-3} threshold has substantial slack relative to both quantities. This is a numerical certificate based on explicit finite inequalities, not an interval-arithmetic proof of every floating-point operation. Independent policy recursions, kernel comparisons, and direct query solves test the implementation of the inequalities.

S.7.4. *Whole-Interval Action Separation*

At a fixed date and state, choose one feasible first action a and one library continuation p_i . Both its reward and its continuation value are affine in the contract. For each competing first action b , the upper-chord continuation gives another affine function. If their difference is positive at both interval endpoints, it is positive at every convex combination of the endpoints. Taking a supremum over all competing actions then proves the main action-separation proposition.

The implementation maximizes the minimum of the two endpoint margins over candidate pairs (a, p_i) . This ordering of maximization and minimization is essential. Taking the minimum of two independently maximized endpoint margins would permit a different favorable witness at each endpoint and would not establish a single affine lower bound on the interval. Saved output includes the selected library index, the first action, both sign margins, and the reported status of each interval. The displayed negative-region endpoint is rounded downward and the positive-region endpoint upward; neither rounded interval extends beyond its certificate.

S.7.5. *Relation to Existing Policy-Transfer Results*

The feature decomposition and the policy-improvement logic build on [Barreto et al. \(2017\)](#), [Chang \(2021\)](#). The anchor upper bound is the convex-combination case of [Nemecsek and Parr \(2021\)](#), whose policy-cache method already addresses whether an existing policy is sufficiently accurate for a new reward function. Our construction uses that bound on the economically reduced stopped-contract parameter, checks every interval breakpoint and every finite state-date node, and applies a separate interval-wide action-separation inequality. The independently verified finite-economy results and the stopping-contract interpretation are the substantive objects reported here. We do not claim that the underlying supremum-of-affine-functions argument, policy caching, or actor-critic separation was first introduced in this paper.

S.8. ADDITIONAL COUNTERFACTUAL AND EQUILIBRIUM COMPUTATIONS

S.8.1. *Joint Refinement and Contract Sensitivity*

The complete refinement panel is reported in Table [S.4](#). Unlike refining time on a fixed interpolation grid, these calculations change state spacing, time spacing, and the action mesh together. They retain the consumption and preference controls as active optimization coordinates. The coarse, intermediate, and fine initial values at $d = 0$ are different; the portfolio at the midpoint is also sensitive to the first refinement. The endpoint contract comparison is more stable than the estimated switching location. All entries remain evaluations of declared finite economies rather than upper and lower bounds on the stopped diffusion.

The correlation panel in the machine-readable output independently resolves the intermediate economy at correlations -0.25 , 0 , and 0.25 . For each, the initial share

TABLE S.4
JOINT TIME, STATE, AND ACTION REFINEMENTS

State grid	Dates	Action grid	d	Value	Duration	π_0
17×25	4	$5 \times 5 \times 7$	0.0	-1.140548	0.884705	-0.50
17×25	4	$5 \times 5 \times 7$	0.5	-0.698148	0.885220	-0.50
17×25	4	$5 \times 5 \times 7$	1.0	-0.253215	0.891936	0.80
33×49	8	$7 \times 7 \times 11$	0.0	-1.136633	0.887830	-0.50
33×49	8	$7 \times 7 \times 11$	0.5	-0.692395	0.890237	0.80
33×49	8	$7 \times 7 \times 11$	1.0	-0.246813	0.894453	0.80
49×73	16	$9 \times 9 \times 13$	0.0	-1.133946	0.888358	-0.50
49×73	16	$9 \times 9 \times 13$	0.5	-0.689493	0.890188	0.80
49×73	16	$9 \times 9 \times 13$	1.0	-0.243752	0.894848	0.80

All controls are reoptimized at every resolution. These refinements use the displayed meshes without the extra frozen neural proposals. The endpoint reversal survives, whereas the midpoint changes sign between the first and second resolution. This table measures sensitivity of finite economies, not a certified bound for the continuous stopped diffusion.

TABLE S.5
ADJUSTMENT QUANTITY AND COEFFICIENT-WEIGHTED EXPENDITURE

k	C^{p_k}	kC^{p_k}	Value
0.5	0.014614	0.007307	-1.117524
2.0	0.009888	0.019777	-1.135835
8.0	0.001432	0.011455	-1.160291

The complete finite problem is solved separately at each coefficient on a 25×37 grid with eight dates and $9 \times 9 \times 13$ actions. The unweighted quantity is decreasing; the coefficient-weighted charge is not. Stopping is included in both quantities.

changes from -0.5 at $d = 0$ to 0.8 at $d = 1$. We do not transfer the feature bank across these changes in transition law. Duration, values, and all controls are stored for each separate solve. The normalization comparison, by contrast, changes only the reward coefficients and retains the same transition and policy class.

Table S.5 distinguishes discounted adjustment effort from its coefficient-weighted charge. The same decomposition is evaluated at the original cost-panel grid and with the original complete control vector. The theorem applies to the first quantity; the middle coefficient produces the largest weighted charge in this panel.

S.8.2. Matched Actor-Free and Quadratic Comparisons

The nine actor-free comparisons reload the deposited nonlinear critic weights and the same initial states. Both actor-start and zero-start algorithms use the same projected convex improvement method, gradient, Lipschitz step bound, and stopping tolerance. The full policy is then evaluated through all 64 terminal shock paths. Saved output separates actor initialization, optimization time, iterations at each date, complete life-time cost, and the reference gap. The original proposal-only ablation remains in S.6; removing the safeguard and removing the actor are different experiments.

To isolate training costs, critic-only backward fits use the same training-state draws and feature construction as the original nonlinear critic. Actor distillation is timed

TABLE S.6
ISOLATED ACTOR COSTS AND INITIALIZATION SAVINGS

d	Seed	Actor train	Actor query	Zero query	Projected break-even
4	101	1.483	0.0414	0.0424	48512
4	202	0.741	0.0432	0.0451	12544
4	303	0.742	0.0436	0.0429	None
8	101	0.808	0.0607	0.0626	13568
8	202	0.797	0.0637	0.0713	3392
8	303	0.794	0.0489	0.0512	11488
16	101	0.878	0.1004	0.1035	8864
16	202	0.879	0.1174	0.1326	1856
16	303	0.875	0.1051	0.1079	9888

Seconds per 32-state complete-tree query batch. Actor training is the distillation fit alone; common target construction and critic training are not charged twice. The last column is $32 \lceil T_{\text{actor}} / (t_{\text{zero}} - t_{\text{actor}}) \rceil$ when the measured saving is positive. It is a linear fixed-batch projection, not an observed crossover. “None” means that actor initialization was not faster in that timing comparison. Optimizer iteration counts and separated initialization times are saved in the machine-readable results.

separately against the same improved-action targets. Table S.6 reports the complete nine-case timing panel. A positive measured online saving is converted into a clearly labeled fixed-batch break-even projection. The separately executed fresh-state workloads in the main paper, not this projection, support the observed amortization comparison.

The quadratic comparator minimizes squared fitting error over

$$q(x) + b + \ell^T z + z^T H z, \quad z = x - \mathbf{1}, \quad H \succeq 0.$$

The regression is convex in (b, ℓ, H) . Its gradient is computed from the full quadratic design matrix. Projected accelerated gradient leaves (b, ℓ) unrestricted and projects the symmetric matrix onto the positive-semidefinite cone by clipping negative eigenvalues. A small ridge term stabilizes the design. The reported projected-gradient norm is below the declared 10^{-8} stopping threshold in the executed fits. At deployment the analytic gradient is $\nabla q(x) + \ell + 2Hz$, and its Hessian bound is the largest eigenvalue of $\nabla^2 q + 2H$. The common investment projection and full scenario-tree cost calculation are unchanged. This comparison tests a genuinely coupled convex quadratic, rather than a separable or unoptimized baseline.

The fresh-state workload charges only for the computations used by each competing method. The reusable policy’s cost is critic training plus zero-start online improvement and full policy evaluation. The specialized solver’s cost is a complete nonanticipative reference optimization at tolerance 10^{-3} . A reference solved more tightly to gap below 10^{-7} supplies the accuracy upper bounds for both; it is not charged asymmetrically as part of one solver’s online cost. Hardware, dependencies, seeds, and individual times accompany the outputs. A three-period, rank-two-shock experiment cannot establish dimension-independent performance, even when its measured query reuse is useful.

S.8.3. Persistent Capacity and All-Date Dynamic Deviations

Let $b_i = 1 - sK_i$, so that next-period capacity probability is $sK_i + b_i a_i$. Write $W_i(k'_1, k'_2)$ for player i ’s continuation value after averaging over next demand. For player one the

best response is

$$a_1 = \text{clip} \left(\frac{\delta b_1}{\kappa} \{W_1(1,0) - W_1(0,0) + (sK_2 + b_2a_2)\Delta W_1\}, 0, 1 \right), \quad (\text{S.2})$$

where $\Delta W_1 = W_1(1,1) - W_1(1,0) - W_1(0,1) + W_1(0,0)$. The corresponding formula holds for player two. The action criterion is strictly concave because the continuation is affine in the player’s own probability and the action cost has curvature $\kappa > 0$. Enumerating whether each clipped best response is at zero, interior, or one gives nine active-set combinations. Feasible candidates are checked against both original best-response equations.

The product of the two affine slopes bounds the Lipschitz constant of the composed best response; clipping is nonexpansive. Its recorded maximum below one therefore provides a unique one-period fixed point for every continuation game. Backward induction supplies a finite-horizon Markov-perfect profile. This argument does not assume current-capacity independence and does not rely on convergence from a single initialization.

For independent dynamic verification, let $J_{i,n}$ be the value of the frozen profile and define the full unilateral best-response value

$$B_{i,n}(s) = \max_{a_i} \{r_i(s, a_i, p_{-i,n}(s)) + \delta \mathbb{E}[B_{i,n+1}(s') \mid s, a_i, p_{-i,n}(s)]\}.$$

At *every* backward date, the code stores $B_{i,n}(s) - J_{i,n}(s)$ for both players and all eight states. It reports the largest positive gain, the signed maximum and minimum, and the attaining date, player, capacities, and demand. The historical code had placed the comparison after the date loop, which verified only date zero. The corrected scope is executed for both the renewal regression and the persistent economy.

As a negative control, the final-period investment of player one at capacities (1,1) and low demand is replaced by one. The independent routine recomputes the frozen-profile value and all best responses. It detects a gain exceeding 0.1, at a recorded late-date node. This test would not be passed by a routine that merely reports convergence of the equilibrium solver. The reset case $s = 0$, four dates, reproduces its early closed-form actions and terminal investment to numerical precision. Both examples remain exact finite-state references, not neural-equilibrium training results.

S.8.4. Executable Files and Preservation

The fifth-revision package is in `replication/r5`. It imports the unchanged fourth-revision transition, resource primitives, and saved neural checkpoints. The contract computations are in `contracts.py` and `economic_robustness.py`. The remaining experiments are in `resource_ablation.py` and `persistent_game.py`. The table generator reads executed JSON; the validation script replays policy features, normalization identities, kernel comparisons, and certificate conditions. The build records input and output hashes and preserves the exact review-base commit.

Earlier papers, theorem proofs, raw negative controls, referee reports, and neural checkpoints remain in their original paths. The new manuscript carries the original mathematical and economic extensions forward and adds the contract-reuse and persistent-capacity material. The accompanying response identifies where each issue in the fourth-revision report is treated. Build completion and numerical verification do not assert journal submission or editorial approval.

S.9. TRANSITION-LAW CERTIFICATES AND MECHANISM REPLICATION

S.9.1. Coefficient Construction and Interval Bounds

For a fixed policy, pre-draw the h remaining iid regime labels. Conditional on their count j , labels are exchangeable, the first label is 1 with probability j/h , and the remaining count is $j - 1$ or j . Applying the same count-blind policy to these conditional paths gives the main coefficient recursion. Averaging over the binomial count distribution proves the polynomial representation. This is an evaluation identity, not a policy allowed to condition its action on the count.

For the separate upper calculation the controller is allowed to observe the remaining count, and past labels are revealed after their transitions. Every original policy can ignore this extra information. Backward maximization of the weighted common-action criterion therefore gives an upper coefficient vector. Averaging over counts preserves the upper inequality. Optimizing the regime-0 and regime-1 actions separately *before* taking their weighted mean would instead disclose the current label and define a different relaxation. The implementation takes the maximum after the mean.

If $h \geq 2$, the degree- h Bernstein coefficients of the corrected upper value on a local interval are

$$U_{n,j} = \left(1 - \frac{j}{h}\right) v_n^a + \frac{j}{h} v_n^b + \frac{j(h-j)}{h(h-1)} M_n, \quad j = 0, \dots, h.$$

At $h = 1$ the correction is zero because the common terminal payoff has zero endpoint difference; at $h = 0$ the only coefficient is g . Degree elevation thus represents the quadratic upper function exactly. De Casteljau subdivision restricts both upper and lower coefficients to the same closed cell before taking their coefficient differences. The bound is conservative because $\sup_t \min_p f_p(t) \leq \min_p \sup_t f_p(t)$. The minimum is taken separately for each state and date, not before the statewise maximum.

The stopping condition uses the maximum of these cell bounds, not the largest loss observed at replay parameters. The replay additionally checks the whole value vector, dominance by the upper function, and improvement from switching relative to the best stored fixed policy. The script stores each anchor policy, its independently evaluated polynomial coefficients, each interval's upper coefficients, cell bounds, and validation locations. The count-information and corrected-chord outputs remain separate so that a loose information bound cannot be substituted for the stopping certificate.

S.9.2. A Reusable QP with the Original Information Structure

At a tree node, write its state as $x_h = M_h x + S_h z + b_h$. Let $Q = I/d + 0.211^\top/d^2$ and let p_h be its probability. Expansion of all stage and terminal costs gives

$$H = 2 \sum_h p_h S_h^\top Q S_h + 2 \text{diag}_h(0.1 p_h I/d), \quad F = 2 \sum_h p_h S_h^\top Q M_h, \quad f_0 = 2 \sum_h p_h S_h^\top Q (b_h - \mathbf{1}).$$

The control diagonal includes decision nodes only. The sums over state costs include terminal nodes. Let D repeat each decision node's probability across its d controls. A Lipschitz constant for the transformed gradient is the largest eigenvalue of $D^{-1/2} H D^{-1/2}$. Each accelerated step is a nodewise simplex projection of $z - D^{-1} \nabla C_x(z)/L$. Every ten iterations we check the complete convex tangent gap; the final gap is independently recalculated by the original tree-gradient routine. The reported query time includes that

evaluation. The structural setup and the learned training times are both included in their respective totals; the tighter deposited reference enters only the accuracy comparison.

The assembly is a new source implementation of the structural comparison described in the fifth-round report, not an invention of parametric quadratic programming. All cold-start workloads use the same saved initial states. No array of known query solutions initializes a new query. Complete costs, gaps, timing samples, feasibility errors, and gradient comparisons are retained in the execution record.

S.9.3. Risk-Class Evaluation and Scope

For a given (d, k) , first solve the original finite problem backward. At the initial state, optimize the current action separately over positive and nonpositive portfolio positions using the same optimal continuation. Evaluate each resulting complete policy independently to obtain (B, A, C) . This computes first-risk-class values, not policies constrained to keep the same sign forever. Repeat the entire backward solve with $\theta = 0$ at every state and date for the no-deliberate-adjustment model. Preference volatility remains present. The two risk-specific option values are nonnegative by set inclusion, and their difference is checked against the independently formed difference of risk advantages.

Bisection starts only when the risk advantage has opposite endpoint signs. The output records both signs, the bracket, the active class features, and the ratio of their effort and duration differences. It does not assign a global uniqueness flag. A duration-gap lower bound valid throughout an interval would justify the main uniqueness result; the present numerical brackets do not assert that additional hypothesis. The portfolio-sign exclusions in the inherited reward experiment therefore retain their original independent status.

S.9.4. Evidence and Reproduction

The current source record is `ECTA_R6.tex`, `SUPP_R6.tex`, the sixth-revision section directory, and `replication/r6/`. Sections S.1–S.8 preserve earlier substantive proofs, specifications, and evidence with their historical identities. The new manifest pins the reachable source commit and hashes the executable sources, immutable input checkpoints, generated records, and tables. The public execution log distinguishes source preparation, experiments, validation, and document build.

The support comparison uses `bank_comparison.py`. Transition calculations use `transport.py` and `kernel_certificate.py`. The remaining programs are `mechanism.py`, `structural_qp.py`, and the independent small-model checks in `unit_tests.py`. Numerical libraries are single-threaded for timing. Two developmental transition-certificate runs were stopped to eliminate repeated interval calculations and then to replace array multiplication by an equivalent CSR multiplication. The final runs use the cached intervals and verified CSR rows. This is an execution optimization, not a change in the state, action, transition, or welfare target. The all-state scalar-OLS comparison retains the inherited common oracle implementation for both objectives.

The revised source does not reclassify the historical neural fits, exact grid repairs, or finite games. In particular, removal of neural proposals still meets the original reward-bank welfare target, and the reusable QP still supplies the relevant stronger resource comparator. A completed build or a passing finite-model validation does not establish journal acceptance, broad neural superiority, or a diffusion approximation theorem.

S.10. COMPARATIVE UPPER ORACLES AND TOTAL POLICY-BANK ERROR

This section supplies the additional proofs for the matched comparison and the finite-refinement result. All statements concern the same finite state/action target and a scalar affine mixture of positive substochastic operators. Approximation of a diffusion is a separate question.

S.10.1. *Localized Count Dominance*

Fix an interval $[a, b]$, let $h = N - n$, and suppress its superscript. Write $C_{n,j} = (1 - j/h)v_n^a + (j/h)v_n^b$ and $f_{h,j} = j(h - j)/(h(h - 1))$ for $h \geq 2$. The corrected upper coefficients are $U_{n,j} = C_{n,j} + f_{h,j}M_n$. At $h = 0$, both constructions equal g . At $h = 1$, the correction is zero because both endpoint terminal functions are g , and the pure endpoint Bellman backups are no greater than their endpoint supersolutions.

Suppose $W_{n+1,j} \leq U_{n+1,j}$. Positivity permits substitution of U for W in the count recursion. For an interior count, expanding the part containing C gives

$$\begin{aligned} (1 - j/h)T_{n,a}^u v_{n+1}^a + (j/h)T_{n,b}^u v_{n+1}^b \\ + f_{h,j}(K_{n,a}^u - K_{n,b}^u)(v_{n+1}^b - v_{n+1}^a) \leq C_{n,j} + f_{h,j}D_n^u. \end{aligned}$$

This identity includes the reward terms, whose weights sum to one. For $h > 2$, the propagated correction is

$$f_{h,j} \left\{ \frac{h-j-1}{h-2} K_{n,a}^u M_{n+1} + \frac{j-1}{h-2} K_{n,b}^u M_{n+1} \right\}.$$

The two weights are nonnegative and sum to one. Keeping the current action u fixed, the entire expression is at most

$$C_{n,j} + f_{h,j}\{D_n^u + \max(K_{n,a}^u M_{n+1}, K_{n,b}^u M_{n+1})\} \leq C_{n,j} + f_{h,j}M_n.$$

For $h = 2$ the next correction is zero and the same conclusion follows from D_n^u . Counts zero and h use the endpoint supersolution inequalities directly. Maximization over the common action completes induction. This is the comparison identified in the R6 advisory report, including its same-action and short-horizon qualifications.

For the rectangular comparison, induction gives $W_{n+1,j} \leq B_{n+1}$. Each count backup is then a convex combination of two quantities no greater than $\max_{u,e} T_{n,e}^u B_{n+1} = B_n$, so $W_{n,j} \leq B_n$. Each original policy can ignore count information, proving that the count polynomial itself is an upper value. Positive polynomial evaluation and positive de Casteljau subdivision preserve coefficient dominance. Subtracting the same lower coefficients, taking coefficient maxima, policy minima, positive parts, and state/date maxima preserves the ordering of the certificates. These statements do not establish that count is faster.

S.10.2. *Operation Counts and Storage*

With exact endpoint values shared, count coefficients $j = 0, h$ need no additional Bellman solve. At remaining horizon $h \geq 2$, all h next-date coefficients require one multiplication by each base kernel. Neighboring results can be cached while looping through j . The total is $2 \sum_{h=2}^H h = H(H+1) - 2$ action-wide kernel applications. The

correction has one two-application nonzero layer at $h = 2$ and four applications at each later layer, totaling $4H - 6$ for $H \geq 2$; it is zero for $H = 1$. The rectangular recursion uses two applications per date. Reward combinations and maxima are additional vector operations. The production code's counts are checked by an instrumented dense backend at horizons 1, 2, 3, 4, 6, 8, 12, 16.

For S states, the correction stores $H - 1$ nonzero state vectors; known zero vectors are not retained in its compact representation. The count method has $\sum_{h=1}^H (h - 1) = H(H - 1)/2$ interior coefficient vectors. Endpoint values are common inputs. The coefficient bank for one feasible policy contains $\sum_{h=0}^H (h + 1)$ state vectors; across m policies this is $O(mSH^2)$, in addition to $O(mSH)$ policy indices. The implementation expands the chord one date at a time during certification, and charges that expansion to certificate time. These persistent-array counts exclude action-wide work buffers and the sparse kernel, and are not claims about process peak resident memory. If a row has at most z stored successors and A actions, an action-wide application costs $O(SAz)$. Naive repeated de Casteljau restriction of m_l local policy arrays to B cells costs $O(Bm_lSH^3)$; it can dominate when action optimization is cheap.

S.10.3. Proof of the Total Coefficient-Error Bound

Let V_n, L_n, Q_n be the primitive recursions in the main text. By backward induction, every fixed-policy value and every optimal value has absolute magnitude at most V_n . For any bounded continuation f with $\|f\|_\infty \leq V_{n+1}$,

$$\|T_{n,\lambda}^u f - T_{n,\lambda'}^u f\|_\infty \leq |\lambda - \lambda'|(\rho_n + \kappa_n V_{n+1}).$$

Monotonicity and the row-mass bound give a Lipschitz constant L_n for both optimal values and all fixed-policy values, whether or not the maximizing action changes.

A coefficient bound is needed in addition to this pointwise result. Let p^a be an endpoint-optimal Markov policy and let $F_{n,j}^{p^a,I}$ be its Bernstein coefficients on $I = [a, b]$. At date n , its conditional-count recursion is a convex combination of endpoint- a and endpoint- b backups of next-date coefficients. Relative to the pure- a policy value, induction and the preceding inequality give

$$\|F_{n,j}^{p^a,I} - J_n^{p^a,a}\|_\infty \leq \beta_n L_{n+1} w + (\rho_n + \kappa_n V_{n+1}) w = L_n w.$$

The same induction bounds the absolute coefficient magnitude by V_n . It is valid even though conditional counts change the regime probabilities: the current mixture weights are between zero and one, and the induction bound is uniform over the remaining count. Exact endpoint optimization gives $J_n^{p^a,a} = v_n^a$ at every state and date.

Next, $\|v_n^b - v_n^a\|_\infty \leq L_n w$, while the signed cross term obeys $\|D_n^u\|_\infty \leq \kappa_n L_{n+1} w^2$. The correction recursion therefore implies $\|M_n\|_\infty \leq Q_n w^2$. For $h \geq 2$, $0 \leq f_{h,j} \leq 1/2$. Consequently,

$$U_{n,j}^I - F_{n,j}^{p^a,I} \leq 2L_n w + \frac{1}{2} Q_n w^2.$$

For $h = 1$ the correction is zero, and at the terminal date the difference is zero. Selecting this one feasible policy already establishes the claimed bound; taking the better endpoint policy can only tighten it. Subdivision is positive and cannot invalidate the coefficient bound. Count dominance transfers the bound to the local count construction.

With $w \leq \epsilon/(4L_*)$ and $w \leq \sqrt{\epsilon/Q_*}$, the linear and quadratic terms are each at most $\epsilon/2$. A complete dyadic grid first reaching width w_* has fewer than $2/w_*$ intervals and at most one more anchor. This proves finite termination and the worst-case $O(\epsilon^{-1})$ statement for fixed primitive constants. The result does not require second-order smoothness of optimal values. For nonexact endpoints, their independently certified one-sided residual and policy-evaluation errors must be added before this termination rule is used.

S.10.4. *Scope of the Matched Frontier*

The full experiment reoptimizes every original R6 transition anchor on the unchanged target and independently reevaluates its policy coefficients. It uses the same endpoint-policy pair and eight closed cells on every compared interval. For $d = 0.5$, it also compares common coarser banks retaining every second or fourth original anchor, always including 1. These locations are inherited computational design choices, not newly timed autonomous anchor searches. Setup, anchor optimization, policy evaluation, local coefficient restriction, upper construction, and cell certification are charged to each method's reconstructed total. Shared work is measured once and attributed identically. Numerical ordering checks and replay diagnostics are validation overhead, not part of any arm's deployment workload. The results are one complete serial, single-thread pass per row on one machine; no cross-machine historical time is used as a comparator. The raw stage times and every interval bound are deposited.

S.11. PRIMITIVE ADJUSTMENT BOUNDS AND DECISION-CERTIFICATE REPLICATION

S.11.1. *The Primitive Option Bound*

For $z = u - 1 > 0$ and $a = \log c$,

$$U(c, u) = -\frac{e^{-za}}{z}, \quad U_u(c, u) = \frac{e^{-za}(1 + za)}{z^2}.$$

Differentiating again gives the strictly negative second derivative in the main text. For finite positive consumption support the expectation can be differentiated term by term. Taylor's integral formula and concavity imply, for $|\theta| \leq \bar{\theta}$,

$$g_\sigma \theta - \frac{1}{2} M_\sigma \theta^2 \leq F_\sigma(\theta) - F_\sigma(0) \leq g_\sigma \theta.$$

Subtracting $k\theta^2/2$, evaluating the lower quadratic at $\theta = g_\sigma/(k + M_\sigma)$ when this is feasible, and maximizing the upper quadratic on the whole line give the option bounds. The upper bound remains valid if the actual optimum is constrained. Subtracting the negative-class upper bound from the positive-class lower bound gives the relative-option inequality.

Since $D_A = A_+ - A_- > 0$ is a primitive constant in this specialization,

$$\Delta^0(d) = B_+ - B_- + dD_A, \quad \Delta^{\text{adj}}(d) = \Delta^0(d) + \Omega_+ - \Omega_-.$$

Both functions have the same strictly positive slope. Their unique indifference contracts therefore differ by $(\Omega_+ - \Omega_-)/D_A$. This proves the stated threshold shift and the intervening interval of opposite rankings. Uniqueness here follows from the explicitly constant duration difference in the two-stage model; it is not carried over to the full stopped portfolio model.

S.11.2. A Curvature Bound over an Entire Primitive Family

Curvature does not have to be estimated by sampling preference indices. In the notation above,

$$-U_{uu}(c, 1+z) = \frac{e^{-za}[(1+za)^2 + 1]}{z^3} = \int_a^\infty t^2 e^{-zt} dt.$$

Its second derivative with respect to z is $\int_a^\infty t^4 e^{-zt} dt \geq 0$. The integral is finite for every $z > 0$, also when $a < 0$. Hence the negative second derivative of expected utility is convex in z , and its maximum on the preference interval is attained at an endpoint. It is affine in the probability of a two-point consumption lottery. Its maximum over the stated probability/preference rectangle can therefore be bounded at its four vertices.

With $u_0 = 2$, $\bar{\theta} = 0.2$, $C_- = 0.5$, and $C_+ \in \{0.05, 0.8\}$, the positive-class shadow is affine and negative for every $p \in [0.06, 0.10]$. Its absolute value lies between 1.4820 and 3.1180; the absolute negative-class shadow is below 0.6138; and $M_+ < 17.342$. Feasibility of the lower-bound trial follows from $3.1180/20 < 0.2$. For every $k \in [20, 80]$,

$$\Omega_+ - \Omega_- \geq \frac{1.4820^2}{2(80 + 17.342)} - \frac{0.6138^2}{2 \cdot 20} > 0.00186.$$

The bound is conservative because it takes adverse cost values separately in its positive and negative terms. Division by $D_A = 0.5$ gives a reduction greater than 0.00372. This proves a statement on the entire probability/cost rectangle, not on the finite set of displayed optimizations.

TABLE S.7
PRIMITIVE PREDICTIONS AND OPTIMIZED ADJUSTMENT OPTIONS

p	k	Lower option	Actual option	Min. shift	Test d
0.06	20	0.0261621	0.0311074	0.0523241	0.723838
0.06	40	0.016882	0.0185198	0.0337641	0.733118
0.06	80	0.00973227	0.0102147	0.0194645	0.740268
0.08	20	0.0681238	0.0811337	0.136248	1.43188
0.08	40	0.0441692	0.0488284	0.0883384	1.45583
0.08	80	0.0257477	0.0271885	0.0514953	1.47425
0.10	20	0.12072	0.147117	0.24144	2.12928
0.10	40	0.0800377	0.0900327	0.160075	2.16996
0.10	80	0.0475675	0.0507768	0.095135	2.20243

Notes: Two-stage CRRA family: $u_0 = 2$, $|\theta| \leq 0.2$, $C_- = 0.5$, $C_+ \in \{0.05, 0.8\}$, and $A_+ - A_- = 0.5$. Option denotes $\Omega_+ - \Omega_-$. The primitive lower bound determines the test contract, at which every row has $\Delta^0 < 0 < \Delta^{\text{adj}}$. The continuous-family proof covers all $p \in [0.06, 0.10]$ and $k \in [20, 80]$, not just these rows. Values are in fixed cardinal utility and operating-benefit units.

The table compares the primitive lower bound with a one-dimensional strictly concave optimization in the same cardinal utility units. For each row the test contract is selected by $d = d^0 - \underline{\Omega}/(2D_A)$, before using optimized option values. It therefore tests a prediction from the bound rather than selecting a contract after locating a numerical reversal. The consumption values, probabilities, and duration difference specify an analytical family, not estimated parameters. In particular, the eight-date stopped economy retains its original state dynamics and is not replaced by this specialization.

S.11.3. *The Full-Model Joint-Region Calculation*

The risk certificate uses all 1,617 states, all eight decision dates, and the original 1,568-action target, comprising the common union mesh and the three frozen neural proposals. The declared initial state is $(u, X) = (2, 1.25)$, the adjustment cost is $k = 2$, the contract interval is $[0.4, 0.45]$, and the one-step shock-law mixture probability lies in $[0, 0.25]$. This region was selected during model exploration; it is not a preregistered or exhaustive characterization of all reversals.

For each adjustment regime and each of the four parameter corners, one backward optimization yields both first-risk classes: dates after the first use the regime's common feasible menu, and the two first-date maxima enforce positive and nonpositive risky positions separately. This is eight corner optimizations, yielding sixteen feasible class policies. The no-adjustment menu imposes zero deliberate adjustment at every date and state but retains preference shocks. Later portfolio signs remain unrestricted. Endpoint class values feed eight localized count constructions, one for each contract endpoint, class, and adjustment regime. Each corner policy is independently evaluated at reward contrasts zero and one, giving its Bernstein intercept and duration coefficients; this covers arbitrary intermediate d without a new policy solve.

Convexity of the optimal class value in d validates linear interpolation of its upper polynomials, not of its optimal policies. Feasible-policy values are affine in d exactly. The resulting degree-eight-by-degree-one tensor coefficients certify the entire rectangle. Subdivision is available if a sign interval contains zero; acceptance of the original closed rectangle in the deposited execution means no further subdivision is required. Pointwise replay at the corners checks implementations but is not the source of the continuum sign claim. All upper and lower tensor coefficients, policy indices, and policy-feature arrays are deposited, with a coefficient-only replay program that does not call an optimizer.

S.11.4. *Numerical Error Account*

The exact-arithmetic theorem is stated separately from its floating-point implementation. The application uses an a priori round-to-nearest allowance relative to the *stored finite kernel and reward arrays*. It is not a claim of directed-rounding interval arithmetic, and it does not include errors in choosing the state/action grid, constructing the quadrature model, or approximating a diffusion. Those omissions do not change which finite target the certificate concerns.

Every stored continuation row is nonnegative with at most sixteen pre-aggregation entries. Let $\hat{\beta}$ be the largest computed row sum plus 10^{-12} . The program verifies $\hat{\beta} \leq 1 + 2 \cdot 10^{-12}$. This margin dominates summation error for sixteen nonnegative entries of this size. Let R be twice the largest sum of the absolute stored base reward, duration reward at $|d| \leq 1$, and effort reward at $k = 2$, across both common kernels and every neural-proposal kernel. The use of $|d| \leq 1$ covers the intermediate feature calculation at $d = 1$, not merely the reported interval ending at 0.45. Let $G = 2\|g\|_\infty$ and

$$V = (G + HR) \max(1, \hat{\beta})^H, \quad q = 4096(H + 1), \quad \gamma_q = \frac{qu}{1 - qu}, \quad u = 2^{-53}.$$

The per-class allowance used in the calculation is

$$e = 2\gamma_q(1 + 2V + R) \max(1, \hat{\beta})^H. \quad (\text{S.3})$$

Here the operation budget is deliberately conservative. One continuation dot product requires at most sixteen products and fifteen additions; constructing its sparse row requires at most sixteen additions of original entries. Adding the three reward components, mixing endpoint laws, and mixing conditional-count branches takes fewer than forty additional scalar operations per path per date. An endpoint action maximum is nonexpansive in the sup norm: its error is bounded by the largest candidate error, not multiplied by the number of actions. Thus an endpoint solve, a selected-policy coefficient recursion, or a count recursion has fewer than $128H$ serial scalar operations along any contributing path. Computing intercept/duration features from values at zero and one adds one subtraction and one multiply-add per coefficient, with absolute magnitude bounded by $2V$. A degree- H restriction uses at most two triangular de Casteljau passes, with fewer than $6(H+1)^2$ scalar operations per retained coefficient path even if counted without sharing. For $H=8$, these counts, subsequent differences, and coefficient extrema are dominated by $4096(H+1)$. Positive mixing and row-mass stability propagate the absolute bound by at most the displayed mass factor. The factors two in R, G and outside γ_q cover input rounding margins and the addition of separately computed paths. The formula is an absolute-error allowance; signed reward cancellation is not analyzed with a relative-error claim. No overflow or underflow occurs in the recorded value range.

The same allowance is charged separately to the upper and feasible lower construction. A class-value gap therefore includes $2e$, and a class-difference sign interval also includes $2e$. The relative-option conclusion adds the already enlarged difference intervals. The information-relaxation and policy-bank errors are controlled by their computed coefficient gaps, not absorbed into an unexplained floating-point tolerance. Independent selected-kernel replay checks the actual arithmetic discrepancy, but small observed replay errors are not substituted for the a priori allowance. Supplementary JSON records the value and reward magnitudes, row-mass bound, operation budget, and final allowance.

S.11.5. *Reproduction and Preservation*

Run `bash replication/r7/run_all.sh` from the repository root, then compile `ECTA_R7.tex` and `SUPP_R7.tex`. The driver executes the dense independent checks, the primitive family, the joint-region certificate, the full matched upper comparison, and autonomous dyadic refinement. The full matched frontier reports one measured pass; the autonomous comparison starts each arm with a fresh policy cache and includes all rejected parent intervals. All retained R1–R6 numerical files are inputs or historical evidence, not silently regenerated by the R7 driver. A byte-level preservation check covers every inherited R6 file except the explicitly advanced revision index. A source inventory, environment record, complete stdout/stderr logs, generated tables, and compiled manuscript copies accompany the revision.

S.12. DECISION PRIMITIVES, DYNAMIC MECHANISMS, AND STREAMING REPLICATION

S.12.1. *The Finite Transition Protocol*

The stochastic state at a decision date is a node of the declared preference–wealth grid. Controls belong to the union of the meshes $(9, 9, 13)$ and $(7, 7, 11)$ in consumption, adjustment, and risky share, after removing exact duplicate actions. There are 1,565 common actions, including 185 with zero deliberate adjustment. Three frozen feasible

proposals are appended at each state and date. On the 49-node wealth grid these are exactly the deposited proposals. On the 97- and 145-node grids we extend their three control coordinates linearly in wealth, separately at each preference node and date. The convex control box makes the extension feasible. The common-menu experiment excludes all three proposals at every date, keeping precisely the same 1,565 controls on every grid. It never invokes the historical shortcut that also removed one of the meshes.

The conditional one-step law has four equiprobable sign pairs. Its candidate arrival is the current state plus the increment in the main text. The fraction α is the smallest positive fraction at which the straight segment meets the boundary, truncated above at one. For a current boundary node $\alpha = 0$ and liquidation is immediate. A strictly earlier hit, $\alpha < 1$, pays the discounted liquidation value at the actual hit point and contributes no live transition mass. An arrival at $\alpha = 1$ is settled on neighboring nodes. If the arrival is a boundary node, the next decision has zero operating duration and immediately pays liquidation; this has the same calendar-time settlement value. At the terminal date all surviving nodes pay G . These conventions reproduce the inherited transition constructor; no reflected state, wealth injection, or additional independent exit draw is inserted.

For an interior arrival coordinate $y \in [l, r]$, the settlement probabilities at l and r are $(r - y)/(r - l)$ and $(y - l)/(r - l)$. Their mean is y and their variance is $(y - l)(r - y)$. Conditional coordinate independence makes the conditional covariance diagonal. Applying the law of total covariance proves the variance identity in the main text. The identity applies to the live settlement lottery; early-liquidation payoffs remain evaluated at their actual hit points. It supplies an economic reason for examining granularity rather than dismissing its effects as arithmetic noise.

S.12.2. Proof of the Uniform Cardinal-Tilt Bound

Write $F_\sigma(\theta) = \mathbb{E}[U(C_\sigma, u_0 + \theta)]$. The original curvature proof gives $-M_\sigma \leq F''_\sigma \leq 0$ on the adjustment interval. Adding $a\theta$ leaves the curvature unchanged and replaces $g_\sigma = F'_\sigma(0)$ by $g_\sigma + a$. Taylor's integral formula gives

$$(g_\sigma + a)\theta - \frac{1}{2}(k + M_\sigma)\theta^2 \leq F_\sigma(\theta) - F_\sigma(0) + a\theta - \frac{1}{2}k\theta^2 \leq (g_\sigma + a)\theta - \frac{1}{2}k\theta^2.$$

The positive-class lower quadratic is evaluated at $\theta = (g_+ + a)/(k + M_+)$. The negative-class upper quadratic is maximized without imposing the adjustment bound; relaxing that bound preserves the upper inequality. With $A = 0.25$, the rounded constants from S.11 imply

$$|g_+ + a| \geq 1.482 - A, \quad |g_- + a| \leq 0.6138 + A, \quad M_+ < 17.342.$$

The trial is feasible because $(3.118 + A)/20 < 0.2$. Thus, putting $P = (1.482 - A)^2$, $N = (0.6138 + A)^2$, and $M = 17.342$,

$$\Omega_+(a) - \Omega_-(a) \geq \frac{1}{2k} \left\{ P \frac{k}{k + M} - N \right\}.$$

The expression in braces increases with k . At $k = 20$ it is strictly positive. It follows that the last display is bounded below by

$$\frac{P \cdot 20 / (20 + M) - N}{2 \cdot 80} = 0.000417380273806974 \dots$$

All endpoint constants are rounded in the adverse direction. Their validity over the entire probability and preference ranges follows from the affine probability dependence and endpoint-curvature argument in S.11. The proof therefore does not rely on the 45 optimizer checks supplied with this revision. The positive root of $P20/(20 + M) - N = 0$, with A variable, is about 0.2718421; it is a sufficient-radius boundary for this bound, not the sharp economic threshold at every (p, k) .

The unadjusted class difference is affine in the contract with slope $D_A = 0.5$. Tilting by $a(u - u_0)$ leaves both unadjusted two-stage payoffs unchanged. The unique two-stage indifference therefore shifts by $-(\Omega_+(a) - \Omega_-(a))/D_A$. This proves the uniform contract shift. At $a = 1$ the sign can reverse, as the retained counterexample shows. Neither baseline invariance nor this threshold direction is imposed on the dynamic economy, where u continues to move when deliberate adjustment is prohibited.

S.12.3. *Dynamic Feature Moments and the Transition-Law Experiment*

The six features in the main text are evaluated by six separate selected-policy backward recursions. Flow features include the actual discount annuity on each branch. The liquidation feature includes both early exit and terminal liquidation. The reward representation is exact for the declared finite law. For every newly optimized class policy, a second evaluation reconstructs the candidate increment and its interpolation weights directly from the selected controls, rather than reading the stored sparse kernel. The feature-weighted objective is then compared with both the Bellman value and that direct replay at every state and date.

For changes in (s, a, d, k, z) , the fixed-policy effect is a linear combination of unchanged feature moments. The reoptimization remainder follows by adding and subtracting the value of the baseline optimizer under the new reward. For a kernel change, let V^p denote the old policy value and $\tilde{V}^p$ its value under the new kernel and rewards. The difference obeys the exact recursion

$$\tilde{V}_n^p - V_n^p = \tilde{r}_n^p - r_n^p + (\tilde{K}_n^p - K_n^p)V_{n+1}^p + \tilde{K}_n^p(\tilde{V}_{n+1}^p - V_{n+1}^p),$$

with the appropriate terminal difference. Backward substitution gives the fixed-policy transition effect. Adding the separately reoptimized value adds a nonnegative gain within each class. Taking a four-class contrast need not preserve that sign. The zero-covariance experiment sets both conditional laws' correlation to zero and rebuilds their kernels; it does not identify that law with an equal mixture of the two baseline endpoint laws.

The mechanism program executes 36 specification-contract comparisons, each with all four class values, for 144 class solves. It preserves the original $k = 2$, the eight-date calendar, and the complete action menu. The retained interventions comprise common cardinal tilts, the reference-state normalization multiplier, the liquidation multiplier, the operating-benefit contrast, the mixture weight, and the separate zero-covariance law. Tables select readable comparisons; the JSON retains every run, six moments per class, direct-replay discrepancies, baseline-policy counterfactuals, and reoptimization gains. No fitted data, identified causal shares, or two-stage calibration is inferred from these computations.

S.12.4. *Streaming the Upper and Feasible Lower Recursions*

The count recursion at a date uses only the coefficient array at the immediately following date. Its endpoint coefficients are shared with the endpoint optimal values.

The first-date action restriction can be split into the positive and nonpositive classes after their common continuation has been computed. The same observation applies to the corrected chord: later-date corrections are common to the two first-action classes, and only the initial maximum is split. Thus sharing those continuations changes neither admissible actions nor information. The one-period case has only its two endpoint coefficients and is handled directly.

For a degree- h policy value, the coefficient at count j uses only counts j and $j - 1$ at the next date. We therefore retain two adjacent date arrays and discard each completed later-date array. At the initial date, only the declared initial state's coefficients are retained. Two passes, at operating contrasts zero and one, yield the policy's intercept and duration coefficients. Once evaluated, a policy occupies HN_s integer control indices and $2(H + 1)$ floating-point focal coefficients. An m -policy bank therefore need not retain $m(H + 1)(H + 2)N_s/2$ value coefficients for this focal-state task. The rolling temporary coefficient storage is $O(HN_s)$; policy storage remains $O(mHN_s)$. These savings do not give all-state, all-date policy-value access for free. The inherited all-state certificate and its storage account remain separate results.

For a common bank, replacing an upper coefficient array by a smaller valid one cannot worsen the certificate. The exact coefficientwise ordering from S.10 is unchanged by rolling storage or by splitting the first-action mask after common continuation. The new independent dense program tests the paired implementation for both adjustment regimes and both signs, including a one-date horizon. It implements the conditional-count mixture and admissible actions separately from the production routine. Its 60 random models give 240 class recursions and 720 point comparisons. Such checks test the implementation; the ordering theorem and induction establish the general result.

The sparse transition implementation aggregates the same at-most-sixteen pre-aggregation entries per row as the inherited positive interpolation constructor. For large wealth grids it retains disjoint CSR blocks rather than allocating a second full concatenated matrix. This storage change preserves rows and rewards. Each block is checked against a separately computed weighted-gather product before use. The common and appended-action matrices, reward arrays, temporary action arrays, policy indices, and coefficient arrays have distinct roles; their costs are not conflated with the number of upper correction vectors.

S.12.5. Arithmetic and Target Error

The revised certificate uses a per-class allowance of 10^{-7} , deliberately larger than the historical count allowance. As in S.11, its scope is floating-point evaluation relative to the stored finite transition and reward arrays, not errors in constructing a diffusion approximation. The code verifies nonnegative rows, bounded row mass, finite rewards, direct selected-policy replay, and coefficient replay. Supplementary evidence records the arithmetic assumptions and the construction checks. The allowance is not a directed-rounding interval enclosure of transcendental evaluations in the transition constructor.

For completeness, let β be the largest stored row mass enlarged by 10^{-12} , let R be twice the largest sum of absolute reward components over $|d| \leq 1$, and let $G = 2\|g\|_\infty$. Set $B_H = G$, $E_H = M_H = D_H = 0$, and $\gamma_j = ju/(1 - ju)$ for $u = 2^{-53}$. The audit uses

$$\begin{aligned} B_n &= R + \beta B_{n+1}, \\ E_n &= \beta E_{n+1} + \gamma_{4096} \{1 + R + 4\beta(B_{n+1} + E_{n+1})\}, \\ M_n &= 4w\beta B_{n+1} + \beta M_{n+1}, \end{aligned}$$

$$D_n = \beta D_{n+1} + 4w\beta E_{n+1} \\ + \gamma_{4096} \{1 + 8w\beta(B_{n+1} + E_{n+1}) + 4\beta(M_{n+1} + D_{n+1})\}.$$

Here w is the mixture interval width, and the last two quantities are zero at the last decision date, where the endpoint continuation values coincide. B bounds exact endpoint magnitudes, E bounds endpoint and selected-coefficient errors, M bounds correction magnitudes, and D bounds their errors. The term $4w\beta E_{n+1}$ charges the propagated difference of two imperfect endpoints. Nonnegative mixing, nonexpansiveness of maxima, and the absolute dot-product error bound justify the recursions. Each sparse row has at most sixteen original entries; aggregation, reward assembly, law mixing, coefficient mixing, and correction construction have substantially fewer than 4,096 serial scalar operations per date. Signed cancellation is bounded in absolute magnitude, not by a relative error in the final value.

An upper coefficient error is bounded by $E_0 + D_0/2 + \gamma_{32}\{1 + 2(B_0 + E_0) + M_0 + D_0\}$. For the lower coefficients we use $3E_0 + \gamma_{16}(1 + 4B_0)$, enlarged by $\gamma_{4096(H+1)}(1 + 4B_0)$ for the two restriction passes. A final $\gamma_{32}\{1 + 4(B_0 + M_0/2)\}$ covers coefficient differences and extrema. The audit requires the larger resulting per-class bound to be below 10^{-7} before a certificate is accepted. An action maximum introduces no factor equal to the number of actions. These deliberately loose counts cover each contributing arithmetic path and do not replace a signed intermediate by its small realized final value.

A conservative absolute arithmetic budget for the new recursions is provided by the audit routine deposited with the revision. It includes endpoint-value error propagated through the signed cross-operator correction, rather than treating that correction as exact. The coefficient inequalities are enlarged separately on their upper and lower sides. In particular, each difference bound includes 2×10^{-7} . The 10^{-3} generic welfare target is not used to certify the much smaller decision margins. An additional error for another economic target must enter the separate class-value transfer inequality in the main text. Observed wealth-grid movement is evidence about the importance of that additional component, not an upper bound on it.

S.12.6. *Reproduction and Version Boundary*

The full revision entry points are `ECTA_R8.tex` and `SUPP_R8.tex`. Run `bash replication/r8/run_all.sh`, followed by the two documented `latexmk` commands. The readable source, generated tables, complete outputs, execution logs, response to the latest report, and preservation manifest are deposited together. The source manifest records hashes, and the workflow records the exact source commit. The historical main paper, all earlier appendices, numerical countercomparisons, and referee reports are retained. New interpretation and additional evidence are incorporated into the complete manuscript, not supplied only in an author-response file.

S.13. PARTICIPATION, SURRENDER, AND PERMISSION FRONTIERS

This section proves the additional contracting results and specifies the numerical target of their implementation. The finite state space, bounded rewards, eight-interval calendar, first-action classification, and frozen later menus are those of S.12. A surrender action is available only at eligible interior nodes. Its reward is $G - F$ and its live transition row is zero. Boundary discharge continues to pay G . The first interval remains compulsory in every reported position-class experiment.

S.13.1. *Affine Payoffs, Participation, and Procurement*

For a deterministic Markov policy, augment the original action labels by a surrender label. Add an absorbing cemetery state if needed to represent killed probability mass. On reaching the surrender label, pay $G - F$ once and receive zero thereafter. The unaltered terminal payoff is G . Backward substitution in this policy recursion, or direct summation over the finite path space, gives

$$J_p = \mathcal{B}_p + d\mathcal{A}_p - F\mathcal{S}_p.$$

The operating duration is nonnegative and bounded by $(1 - e^{-\rho})/\rho$. An elective surrender occurs at most once, so $0 \leq \mathcal{S}_p \leq 1$. In particular, the fee exposure cannot be replaced by the expected number of visits to surrender-eligible states.

There are finitely many deterministic Markov policies in the original action target. They attain the unrestricted and sign-restricted optima; randomization does not improve a finite-horizon expected payoff. Each policy payoff is affine in (d, F) , so their maximum is convex, nondecreasing in d , and nonincreasing in F . If $F' \geq F$, every fixed policy satisfies $-(F' - F) \leq J_p(F') - J_p(F) \leq 0$. Taking policy maxima on the appropriate side gives the 1-Lipschitz property. Increasing the minimum term removes surrender policies, so it weakly lowers each class value. None of these orders implies an order for a difference of class values.

For completeness, consider a direction $v = (v_d, v_F)$ at a contract $z = (d, F)$. A finite set of nonoptimizing policies has a strictly positive value gap from the maximum, unless it is empty. For sufficiently small positive steps, only policies optimizing at z can determine the directional derivative. The derivative is therefore

$$D_v W(z) = \max_{p \in \arg \max J(z)} \{v_d \mathcal{A}_p - v_F \mathcal{S}_p\}.$$

This proves the exposed-feature statement, including the case of several policies with the same feature vector. It is the standard finite envelope logic used in the main paper, not a new general envelope theorem.

The external numeraire makes total agent utility $q - C(E) + \max_\sigma W_\sigma^r$. Comparing this quantity with $G(x_0)$ and imposing $q \geq 0$ gives the stated participation grant. The managed initial state is unchanged, so the grant does not enter the Bellman recursion. If a principal receives flow b while the mandate operates, its expected payoff at the minimum grant is $b\mathcal{A}_{p_r^*} - q_r^*(E)$. Termination charges go to the separate enforcement institution. Dividing by strictly positive duration gives the procurement condition. If an agent optimum is not unique, the contract must specify an admissible selection or use the relevant worst-case duration; the proposition uses a specified selection. It does not assume that all payoff-equivalent policies deliver identical service.

Both regime policy sets use the same financial opportunities and outside payoff. The no-adjustment policies are feasible in the adjustment regime. Therefore $W_\sigma^{\text{adj}} \geq W_\sigma^0$ and $q_\sigma^{\text{adj}} \leq q_\sigma^0$ at a common capacity cost. Where all class-specific participation gaps are positive, subtracting the two participation equations within each class gives $q_\sigma^0 - q_\sigma^{\text{adj}} = W_\sigma^{\text{adj}} - W_\sigma^0$. The four-class contrast yields the price interpretation of the relative option. The sign of that contrast is not implied by its two component inequalities.

The separate-account assumption is economically material. A model in which all payments enter managed CRRA wealth would require a new state transition, consumption response, and participation calculation. Similarly, collateral opportunity costs

depending on realized duration would be additional running or terminal rewards, rather than the ex ante $C(E)$ used here. These are alternative economic targets, not accounting details that can be inserted without reoptimization.

S.13.2. *The Statewise Enforcement Frontier*

Let R_n be the live support reachable using the union of endpoint kernels and all original feasible actions. Nonnegative kernel entries imply support closure: if $x \in R_n$ and a live row from x gives positive mass to y , then $y \in R_{n+1}$. Elective surrender can remove paths, but cannot introduce a new live state. The union may be larger than the support of any particular optimized policy; this is intentional.

Suppose $F \geq F_r^*(m; \lambda, d)$. At maturity the optional and compulsory values both equal G . Suppose they coincide at all reachable nodes at date $n + 1$. By support closure their operating backups coincide at any reachable node at date n . At an eligible interior node, $G - F \leq V_n^r$ by the definition of F_r^* ; taking the maximum with surrender cannot raise the value. At an ineligible node the operating backup is the only option, and at a boundary the payoffs agree by construction. Backward induction proves statewise equality, and then equality of both sign-restricted initial values. If instead $F < F_r^*$, some eligible reachable interior node has $G - F > V_n^r$, at which its optional value is strictly higher. This proves the necessary direction for equality at every reachable node. Such a node might not affect either initial optimum, which is why the threshold is not described as the smallest initial-sign-preserving fee.

Increasing m removes terms from the maximum. Increasing d weakly raises V_n^r at every state by nonnegative policy duration, so it weakly reduces the maximum. Policy inclusion gives $V_n^{\text{adj}} \geq V_n^0$, and hence $F_{\text{adj}}^* \leq F_0^*$. Thus adjustment weakly reduces the capacity needed to implement the same mandatory operating values. When C is nondecreasing, an available smaller implementing capacity can also weakly reduce its participation cost. This claim concerns implementation capacity, not monotonicity of the initial risk preference.

The regional calculation uses the two no-adjustment noncancellable policies optimized at $(\lambda, d) = (0, 0.4)$ and $(0.25, 0.4)$. Their later-date policies are feasible regardless of the initial class used to store the policy. For each policy and each date $n \geq 1$, compute its Bernstein coefficients in the *physical* mixture probability λ . Restrict those coefficients to each of sixteen subintervals of $[0, 0.25]$. Their coefficient minimum is a uniform policy lower value at every state on that subinterval. Take the larger of the two policy lower values, and subtract 10^{-7} before comparing with G . Since duration is nonnegative, this lower bound also applies for all $d \in [0.4, 0.45]$. It is below V^0 and therefore below both regime values. Maximizing the resulting liquidation deficit over eligible reachable nodes gives the sufficient regional frontier.

The resulting bound 0.8348592805606 for $m = 1$ includes the stated arithmetic allowance. The upward-rounded value 0.834860 is sufficient relative to the stored arrays. The optimizing deficit in this bound occurs at date one, node $(1.95, 1.34375)$, on the first law subinterval. Restricting to the actual initial state alone would not justify ignoring that continuation constraint. Conversely, including every unrelated grid node raises the sufficient bound to 2.5309784144. Neither bound is a claim about a continuum of initially admissible managed states.

Table S.8 gives the direct-solver statewise thresholds at the original center. They use the same reachable support and the separately optimized mandatory continuations. In particular, the one-interval thresholds are approximately 0.7767542 with adjustment and

TABLE S.8
STATEWISE NO-SURRENDER THRESHOLDS AT THE ORIGINAL CENTER

m	Adjustment	No adjustment
1	0.7767542	0.8135332
2	0.7161762	0.7458682
3	0.6442995	0.6661886
4	0.5592377	0.5734029
5	0.4575809	0.4648691
6	0.3346276	0.3367796
7	0.1842712	0.1842768
8	0.0000000	0.0000000

Direct evaluations of $\max_{n \geq m, x \in R_n} [G(x) - V_n^r(x)]_+$ at $(\lambda, d) = (0.125, 0.425)$. Both use the same all-action reachable support. The initial point alone can require a smaller charge than statewise equality. These point calculations are distinct from the upward-rounded sufficient regional bounds in the main paper.

0.8135332 without adjustment. They illustrate the policy-inclusion order. The sufficient regional bounds in the main text additionally cover all contracts in the rectangle and use feasible-policy lower values; the two tables answer different questions.

If G is an operating supersolution at every date, the zero-charge recursion has $W_n = \max(G, \mathcal{T}_n G) = G$ at the last eligible decision date. Repeating the argument proves equality at all eligible dates. Earlier mandatory dates retain their operating recursion. At date zero, admitting free surrender yields $G(x_0)$ as well. In the finite implementation, every operating action is checked at both law endpoints and the upper benefit endpoint. Since the one-step expression is affine in the law weight and its duration coefficient is nonnegative, these endpoint inequalities cover the entire original rectangle. The large observed slack is a supersolution check for the stored arrays, not an interval proof for an uncomputed transition constructor.

S.13.3. Risky-Share Breakpoints and Permission Derivatives

Fix an initial (c, θ) pair. For each law and sign realization the proposed preference coordinate is $2 + h\theta + \sqrt{h}\sigma_u z_u$, independent of π . The wealth coordinate has the form

$$y_X(\pi) = a_c + b_{q,z}\pi, \quad a_c = 1.25 + h(r1.25 - c),$$

$$b_{q,z} = h(\mu - r)1.25 + \sqrt{h}\sigma_X 1.25(qz_u + \sqrt{1 - q^2}z_X).$$

For the declared constants, $|y_u - 2| < 0.043$ and $1.054 < y_X < 1.365$ throughout the wider risky-share domain, all common (c, θ) pairs, and both law endpoints. These conservative bounds lie strictly inside $[1.2, 2.8] \times [0.5, 2]$. The whole line segment therefore stays interior, the exit fraction is one, and the flow annuity is independent of π .

Conditional on a shock, the preference interpolation weights are constant as π varies. Between any two wealth-node crossings, each wealth interpolation weight is affine in y_X , hence affine in π . The bilinear interpolant of any continuation vector is consequently affine in π there. Averaging shocks and the law mixture preserves affinity on the common partition. The flow reward is independent of π , so the first-date action value is continuous and piecewise affine. A maximum over a compact interval is attained at an endpoint or a breakpoint. A positive open interval uses its closure only to compute

the supremum; the implementation reports the chosen action and does not call a zero endpoint a positive maximizer. The original finite frozen proposals are compared separately when feasible.

For an interpolation cell, write $I_X V_1(y)$ for the slope of the interpolant in wealth. Differentiation gives

$$\partial_\pi Q(c, \theta, \pi) = \sum_q \Pr(q) \frac{1}{4} \sum_z e^{-\rho h} I_X V_1(y_{q,z}(\pi)) b_{q,z}.$$

There is no within-period flow derivative in this expression. If the winning pair is locally fixed and the relevant optimum remains at L or $-S$, the value of relaxing the permission is the corresponding endpoint derivative. At a breakpoint or an action tie, use the appropriate one-sided piecewise derivative or directional envelope. Because the first-date permissions do not change V_1 , no later-menu derivative is being omitted.

Changing d , unlike changing only a first-date permission, changes the optimized continuation. The finite envelope theorem nevertheless gives the derivative of each class value as its optimizing discounted duration. Differentiating $\Delta'(\lambda, d_r^*, F, L, S) = 0$ at a regular crossing then gives the three frontier formulas in the main text. If $D_A^r = 0$, if a feature vector is not exposed, or if a control switch destroys differentiability, these ratios are not asserted. The deposited benefit-root calculations reoptimize continuation at every trial benefit and retain local endpoint signs. Their floating-point root brackets are not directed-rounding interval enclosures and do not prove that no other crossing exists.

S.13.4. *The Joint Fee–Benefit–Law Certificate*

A fixed stopping policy evaluated under affine endpoint kernels has a polynomial value of degree at most the remaining horizon in λ . This follows directly from the recursion: each live step multiplies by one affine mixture, whereas a surrender row has zero live kernel. Padding with zero future rewards and a cemetery state permits degree elevation to the common horizon. Applying the same recursion to the base reward, duration reward, and unit surrender charge yields the three focal coefficient vectors B_p, A_p, S_p .

At each (d, F) corner, the signed correction and the count recursion use the operating and surrender actions together. Their proofs in S.10 apply because the added action has a bounded affine reward and a nonnegative zero kernel. The last-decision endpoint continuations agree at the terminal vector, and the initial sign mask is applied only when forming the date-zero class upper values. The count information and the signed correction are not granted a different surrender date or a different charge from the feasible lower policies.

For a fixed law probability, the optimized value is a maximum of affine functions of (d, F) . Its value inside a rectangle is therefore bounded above by the bilinear convex combination of its values at the four corners. Replacing those corner values by valid law-polynomial uppers preserves the inequality. A fixed policy value is affine in (d, F) and equals the corresponding corner interpolation. Restrict both the upper and policy polynomials to a law subinterval and express them in a common Bernstein degree. For a positive policy p and nonpositive upper U_- , the minimum of all coefficients of $J_p - U_-$, including the reward-parameter corner indices, bounds that difference everywhere in the cell. Taking the largest such bound across positive policies remains valid. The upper

difference is obtained symmetrically from $U_+ - J_q$ and a minimum across nonpositive policies. Taking the worst cell gives the regional statement. No policy is selected using information unavailable in its actual evaluation.

The deposit constructs the full four-corner fee-benefit bank at the two law endpoints for both adjustment regimes and both initial classes. The $F \in [0.85, 0.90]$ certificate uses four law subcells, while retaining the original full menu plus the eligible surrender action. The favorable signs are not inferred by extrapolating the pointwise no-surrender observation at the center. The reachable enforcement bound supplies an additional, separate sufficient explanation for their equality with the noncancellable signs. The $F = 0.80$ certificate fails; direct reoptimization at $(\lambda, d) = (0, 0.4)$ gives a positive no-adjustment difference 0.0013923796303, so this failure is not merely an uninformative upper bound.

S.13.5. Arithmetic, Target Transfer, and Executed Validation

The arithmetic assumptions are those in S.12: stored reward arrays and nonnegative transition arrays are evaluated in round-to-nearest binary64 arithmetic. Rewards can have either sign; nonnegativity applies to transition weights and duration, not to utility. The surrender reward is included in the absolute reward bound and its row mass is zero. Let $\gamma_j = j2^{-53}/(1 - j2^{-53})$. The R9 audit reuses the B, E, M, D backward recursions in S.12 with row-mass bound $\beta = 0.9950124791936824$, absolute reward bound $R = 27.2130993863$, and law width $1/4$. The bound covers $|d| \leq 1$ and $0 \leq F \leq 1$.

For the additional three-feature lower representation, the audit uses

$$L_{\text{err}} = 5E_0 + \gamma_{4096(8+1)}(1 + 8B_0),$$

and the same upper error as in S.12, enlarged finally by $\gamma_{64}(1 + 4B_0 + 2M_0)$ for coefficient subtraction and extrema. The factor five charges the subtraction of the base value in constructing the duration and surrender features and their recombination, instead of treating those features as exact. The restriction budget dominates both de Casteljau passes. A sparse row has at most sixteen entries before aggregation; adding one surrender label does not increase a dot-product length. Maxima are nonexpansive and introduce no factor equal to the number of feasible actions. The resulting derived per-class bound is 1.4305784×10^{-8} , below the deliberately larger accepted allowance 10^{-7} . Every displayed difference interval is enlarged by 2×10^{-7} .

The stored-array target is not silently identified with a real-arithmetic transition constructor. For two finite targets with a common action correspondence, subtract their Bellman backups, add and subtract one kernel applied to the other's continuation, and use $|\max_a q_a - \max_a \tilde{q}_a| \leq \max_a |q_a - \tilde{q}_a|$. An absolute reward discrepancy ϵ_r , row- ℓ^1 discrepancy ϵ_K , continuation magnitude bound B , and row-mass bound β give $\epsilon_n \leq \epsilon_{r,n} + \epsilon_{K,n}B_{n+1} + \beta_n\epsilon_{n+1}$. The terminal discrepancy initializes this recursion. Surrender, forced-discharge, and boundary payments must enter the reward or terminal discrepancy. Different action sets require a separate action-approximation bound or a correspondence with a priced improvement error. A diffusion additionally requires its own consistency and approximation argument. The arithmetic allowance already charged to the stored-array certificate is not counted a second time as a measured constructor error.

The new independent validation has two components. Twelve random two-state, three-interval models are checked by enumeration of 81 deterministic continuation policies per initial class and parameter point. There are 120 class-point comparisons.

The largest discrepancy between enumeration and the production stopping recursion is 4.45×10^{-16} ; count never exceeds the signed chord beyond the declared numerical tolerance. A separate full-economy check evaluates 80 class-parameter comparisons across four fees, both regimes, and five mixture probabilities. Selected-control reconstruction without the stored sparse rewards and kernels differs from the optimized value by at most 4.45×10^{-16} . Fixed-policy polynomial replay differs by at most 4.45×10^{-16} . Additional checks cover reward-parameter convexity, action eligibility, and the distinct date-zero free-surrender contract. These are implementation checks, not substitutes for the preceding proofs.

The complete economic run retains 19 center surrender/term specifications, 20 further fee-contract rows, their four class policies and moments, sixteen all-action supersolution checks, the eight sufficient enforcement bounds, nine two-regime permission comparisons, and eighteen reoptimized benefit roots. Three regional fee specifications are checked with both streamed upper constructions. All 398 protected historical or numerical source entries checked by the execution manifest remain unchanged. The permission target is the continuous first-date risky-share envelope conditional on finite consumption-adjustment pairs; the fee certificate target is the original finite menu plus surrender. The deposit does not claim a joint theorem simultaneously enlarging those two different action targets.

S.13.6. *Reproduction*

The complete manuscript entry points are `ECTA_R9.tex` and `SUPP_R9.tex`. From the repository root, run the following commands in order:

```
python replication/r9/run.py -mode all
python replication/r9/validate.py
python replication/r9/tables.py
```

The source inventory, historical hash checks, generated JSON, policy and coefficient arrays, execution transcript, manuscript build log, and point-by-point response are deposited together. R9 adds new readable source files and new evidence directories. The original R8 manuscript, supplement, all earlier economic applications and adverse comparisons, and the latest advisory report remain available at their original paths.

S.14. DYNAMIC TRANSPORT, JOINT CONTRACTS, AND PRICED IMPLEMENTATION

This section supplies the proofs and numerical specifications for the section entitled “Dynamic Adjustment and the Price of Commitment” in the main paper. Main-paper labels in this independently compiled supplement are identified by their proposition titles; the substantive definitions are repeated here to avoid dependence on cross-document references.

S.14.1. *Proof of the Reachable-State Transport Proposition*

Let $V_n^\uparrow$ be the value with nonnegative deliberate drift, including the same elective stop action as the unrestricted problem. For a negative action a , let a^0 be its feasible zero-drift counterpart. Given enclosures $\ell \leq V_{n+1}^\uparrow \leq h$, write $V_{n+1}^\uparrow = v + z$, where $v = (h + \ell)/2$ and $|z| \leq e = (h - \ell)/2$. Entrywise positivity gives

$$(K^a - K^{a^0})V_{n+1}^\uparrow \leq (K^a - K^{a^0})v + |K^a - K^{a^0}|e.$$

For each endpoint regime, maximize the reward difference over the two benefit endpoints. Taking the convex combination of these endpoint upper bounds bounds the difference under every mixture law. Maximizing that affine upper expression over the two law endpoints is valid. This is slightly more conservative than evaluating the absolute value of the mixed signed kernel itself, since $|(1 - \lambda)D_0 + \lambda D_1| \leq (1 - \lambda)|D_0| + \lambda|D_1|$. The charged evaluation allowance then yields $Q^a(V_{n+1}^\uparrow) < Q^{a^0}(V_{n+1}^\uparrow)$.

At terminal time the unrestricted and restricted values agree. Suppose they agree at date $n + 1$ on the common reachable set. Closure of the set under all feasible live transitions permits replacement of unrestricted continuation by restricted continuation in every date- n backup. Every negative-drift backup is strictly dominated by a feasible zero-drift backup. All remaining operating and stop actions are available in the upward problem. Hence the two maxima agree. Backward induction proves equality on every reachable set, including either initial financial class: the counterpart has the same risky share, so it respects the class restriction. Boundary nodes have the common discharge value and require no strict operating-action inequality. The argument uses all feasible successors, not only successors of the ultimately chosen policy.

For completeness, the continuation enclosures used in the computation follow from a separate backward comparison. Start with the common terminal array. At each later date, evaluate the lower continuation at the lower benefit and higher fee and minimize over the law endpoints before maximizing over upward-feasible actions. This is a lower bound, because it permits an adverse law choice for each action. Evaluate the upper continuation at the higher benefit and lower fee and maximize over both law endpoints and actions. This is an upper bound. Add outward error padding after each stage. Neither recursion assumes that the optimal policy is constant on the box.

At the first date, the restricted objective is piecewise affine in the risky share for each fixed consumption–drift pair. The difference between the negative-drift action and its counterpart is piecewise affine on the union of their breakpoint sets. Its maximum over every admissible permission interval is therefore covered by the union of these knots and the largest-permission endpoints. A separately proposed action needs its own counterpart or direct domination proof. The deposited frozen proposals all have nonnegative drift on the reachable live states; this statement is checked rather than assumed.

S.14.2. Continuation Incentives and the Dynamic Class Differential

In a branch on which the exit fraction α is independent of deliberate preference drift, the arrival preference changes by $\alpha h \theta$, while consumption and the financial transition are fixed. The fundamental theorem of calculus for absolutely continuous, piecewise differentiable continuation gives the transport integral in the main paper. Integrating the quadratic flow cost gives $k\theta^2 \mathbb{E}[(1 - e^{-\rho \alpha h})/\rho]/2$. Immediate preference discharge requires a different branch calculation and is not covered by this smooth-transport interpretation; the signed-row test still applies.

An exact decomposition clarifies the dynamic comparison. Suppose the selected first financial controls (c_σ, π_σ) agree between the adjusted and unadjusted class optima, as checked at the reported central contracts. Write $a_\sigma = (c_\sigma, \theta_\sigma, \pi_\sigma)$ and $a_\sigma^0 = (c_\sigma, 0, \pi_\sigma)$. Then

$$W_\sigma^{\text{adj}} - W_\sigma^0 = Q^{a_\sigma}(V_1^{\text{adj}}) - Q^{a_\sigma^0}(V_1^{\text{adj}}) + K^{a_\sigma^0}(V_1^{\text{adj}} - V_1^0). \quad (\text{S.4})$$

The last term measures exposure to future adjustment opportunities under the zero-drift initial action. Subtracting the two classes yields the reported relative local and future terms. If the financial controls differ, the additional term $Q^{a_0}_\sigma(V_1^0) - W_\sigma^0$ must be retained; it is not generally zero. The replication code fails rather than silently suppressing that term when its equality premise fails.

For the boundary-state diagnostic, continuation is further written as $V_1 = G + (V_1 - G)$. This separates the liquidation benchmark from the liability of remaining in operation. The code independently reconstructs noncost flow, adjustment cost, immediate discharge, live liquidation-benchmark exposure, and continuation excess. The lower-preference-boundary mass is obtained by evaluating the boundary indicator through the stipulated first-date settlement lottery. It is discounted incidence, not an undiscounted probability and not a continuous-time hitting probability. The large full-minus-upward value difference is a reoptimized all-date comparison; the one-step component is not substituted for that full difference.

The primitive two-stage result and its common-cardinal-tilt robustness remain separate theorems. No step of this proof substitutes their cost coefficients or adjustment directions for those of the eight-date economy.

S.14.3. *Proof of the Joint-Region Certificate*

Fix a deterministic feasible stopping policy for the dates after the first, and a feasible first-date action. Affine dependence of rewards and kernels on the repeatedly drawn regime gives a degree-at-most-eight Bernstein polynomial in λ by backward conditioning on the number of regime-1 draws. Conditional on the policy, the coefficients are affine in d and F . The first-date coefficient is obtained by applying its selected first-date kernel to the seven-period continuation coefficients. Early stopping shortens effective degree but does not alter the degree-elevated representation.

At every benefit–fee corner, compute endpoint optima for the largest permissions. The signed corrected chord, or the count-information recursion with the same information timing, bounds each class value over the law interval. Convexity of optimized value in (d, F) then bounds an interior reward point by the multilinear interpolation of these corner upper values. Both methods allow the same elective stop action at the same nodes. The first-date upper backup optimizes over the breakpoint menu corresponding to the largest permissions; the later upper backups use exactly the original full action menu.

For lower values, retain feasible policies optimized at the smallest permissions. They remain feasible at every permission pair in the box. Fix one such policy across all reward corners before taking a regional bound. Let $L_{c,j}^p$ be its restricted Bernstein coefficient at reward corner c and law coefficient j , and let $U_{c,j}^\sigma$ be an upper coefficient for a comparison class. Nonnegativity and unit sum of the tensor interpolation weights imply

$$J_p - W_\sigma \geq \min_{c,j} \{L_{c,j}^p - U_{c,j}^\sigma\}.$$

Taking the maximum of this bound over feasible lower policies is valid. Reversing the roles supplies an upper difference bound. Importantly, taking a separate best lower policy at each coefficient or reward corner before the minimum is not justified and is not done. The policy identity is held fixed through the complete corner-and-coefficient minimization.

For the adjusted and zero-adjustment spreads, apply this construction within each regime. For active surrender, apply it to the positive zero-adjustment class with surrender and to that same class with all eight intervals compulsory. The first-date permission extremes are treated in the same conservative manner. The resulting bounds include 2×10^{-7} for each difference. Their displayed decimal bounds in the main paper are rounded outward.

If a policy has zero discounted elective-surrender incidence, its operating behavior and payoff are feasible in the noncancellable economy. It cannot exceed the latter class supremum. A strictly positive lower bound on the value gain from permitting surrender therefore implies that any policy closer to the surrender value than that bound must have positive elective-surrender incidence. This conclusion remains valid for a class defined by a supremum over strictly positive shares. The closure at zero is used only in the upper search; every retained lower policy has a strictly positive initial share in class +.

The proof is a continuum argument, not an inference from the six interior validation contracts. Those direct reoptimizations test the implementation against the derived bounds. The transport certificate supplies the independent statewise directional statement. Neither a point replay nor a class-value bound alone would supply it.

S.14.4. *Proof of the Priced Implementation Proposition*

For regime r , let V_n^r denote the noncancellable continuation and let $\mathcal{R}_n$ be the feasible-support tube. A fee F implements equality of surrender and noncancellable values on all eligible reachable nodes precisely when

$$F \geq F_r^*(m) := \max_{m \leq n < 8} \max_{x \in \mathcal{R}_n \text{ live}} [G(x) - V_n^r(x)]_+.$$

The empty maximum at $m = 8$ is zero. This is a statewise threshold, not a necessary threshold for matching only the initial value. The earlier proof by backward induction applies without change to the enlarged first-date menu because the reachable tube now contains its successors. Policy inclusion gives $V_n^{\text{adj}} \geq V_n^0$ and hence $F_{\text{adj}}^*(m) \leq F_0^*(m)$.

Conditional on implementing the selected noncancellable allocation, W^r and delivered service A_r do not depend on the chosen instrument. For a given term, nondecreasing capacity cost makes the smallest sufficient capacity optimal. Participation requires $q \geq G(x_0) - W^r + C(E)$ and $q \geq 0$; hence the minimum grant is the stated positive part. The principal pays that grant and direct term cost. Minimizing over the eight terms proves the procurement formula, and comparison with bA_r proves the break-even condition. At the threshold, continued operation must be the specified best-response selection. Requiring strict stopping incentives instead gives the infimum as fees decrease to the threshold; the code separately checks strictly higher fees and reports them as implementation tests, not as the exact continuous optimum.

When $z = G(x_0) - W^r \geq 0$, the grant increment from capacity is exactly $C(E)$. For arbitrary z , monotonicity and unit Lipschitz continuity of $[\cdot]_+$ give $0 \leq [z + C(E)]_+ - [z]_+ \leq C(E)$. Comparing this increment with saved direct term cost proves the general instrument comparison, including the nonbinding-participation case. With $D(m) = \kappa(m - 1)/8$ and binding participation, each term's cost is affine in κ . The lower envelope selects the economically preferred institution; crossings of these affine functions, not the free-term omission, determine its changes.

The illustrative coefficient 0.02 and the values of κ are theoretical primitives, not estimated enforcement costs. The principal's fees are zero on the implemented path, so no hidden fee revenue finances the grant. For active contracts outside this implementation family, any different fee recipient or principal revenue assumption must be entered explicitly.

S.14.5. *Arithmetic, Targets, and Reproduction*

The source fixes 1,617 continuation states, the inherited 1,568 later operating actions, eight dates, and the stored first-date breakpoint rows. The first rows are nonnegative with mass no larger than the inherited discount bound. The stopped reward includes the same boundary payoff and the elective fee. A full array manifest records data types, shapes, and SHA-256 identities for the primitive reward, duration, effort, CSR, and first-date arrays generated by the pinned sources.

The gamma-bound calculation in the R9 arithmetic audit is rerun at the new interval width and fee bound. Its per-class bound is smaller than the charged 10^{-7} . The first-date backup has at most sixteen pre-coalescence transition entries per row. A signed difference of two such rows has at most thirty-two; its absolute-value transport bound is charged a further 2×10^{-7} per comparison. The interval recursion is padded by 10^{-7} at each date. These pads exceed the round-to-nearest operation bound for the stored rewards and bounded eight-step continuation values. Positive interpolation weights and maxima do not amplify a uniform one-sided value error beyond the stated row-mass recursion. The support test uses the nonzero entries of these stored nonnegative kernels.

There is an additional distinction at the first date. The certified operator is the piecewise-affine interpolation of its deposited knot rows for each finite consumption-drift pair. Direct evaluation of the original transition formula at selected caps is checked against this representation; agreement is an implementation test, not an enclosure of the real-arithmetic constructor. Rebuilding with another environment requires checking the recorded array identities or treating the rebuilt arrays as a separately identified target. No constructor or diffusion enclosure is inferred from last-bit agreement.

The supplied runner executes the nine directional replays, the alternative-state witness, the dynamic decomposition, all twelve joint fee-permission interventions, both regional upper methods, the full reachable-state transport test, all eight enforcement alternatives, selected-control reconstructions, and interior-region validations. It saves the actual coefficient arrays and lower policies, not only their final extrema. The common endpoint bill, construction, and validation are charged to each matched upper-method workload. Kernel payload is reported separately from process peak memory, which is not measured. Historical training and full historical timing sweeps are preserved but are not described as newly rerun.

The proof establishes the indicated finite-array economic result. Transferring initial value inequalities to another target requires the additional value-error account stated in the main paper. Transferring directional dominance requires a compatible statewise transport and reachability account. These are different mathematical obligations.

S.15. SERVICE, PROCUREMENT, EXPOSURE, AND WITNESS RECONSTRUCTION

This section supplies the additional arguments used by the R12 main paper. Sections S.1–S.14 above are retained in full. Their references to earlier main-paper statements

refer to the accompanying complete theoretical and computational compendium. References with the R12 labels below refer to the new main paper. The numerical target for this section is the canonical R12 array system; identities of historical targets are not silently reassigned.

S.15.1. Service bounds without differentiability

Fix an item j , an adjustment regime, and a feasible policy π . The definition of operating duration gives

$$J_j(\pi; d + h) = J_j(\pi; d) + hA_j(\pi), \quad J_j(\pi; d - h) = J_j(\pi; d) - hA_j(\pi). \quad (\text{S.5})$$

These equalities hold with surrender, forced discharge, deliberate adjustment, and stochastic kernels. The benefit parameter changes only the reward coefficient, not a kernel, feasible set, capacity cost, term cost, or stopping charge. This is why the two perturbed problems can identify service. Varying a primitive that also changes the transition or stopping technology would not justify the same argument.

PROOF OF PROPOSITION 1: Let π be η -optimal at d . Optimality of the perturbed value and near-optimality at the center imply

$$W_j(d + h) \geq J_j(\pi; d + h) \geq W_j(d) - \eta + hA_j(\pi), \quad (\text{S.6})$$

$$W_j(d - h) \geq J_j(\pi; d - h) \geq W_j(d) - \eta - hA_j(\pi). \quad (\text{S.7})$$

The first inequality yields $hA_j(\pi) \leq W_j(d + h) - W_j(d) + \eta$; the second yields $hA_j(\pi) \geq W_j(d) - W_j(d - h) - \eta$. Substituting the adverse endpoints of the three certified intervals proves (7). Intersecting with $[0, \bar{A}]$ preserves validity. Randomization preserves the two affine identities after expectation, and the same inequalities follow. No limiting derivative or stable optimizer has been assumed. If the positive-class supremum is not attained, every existing feasible near-optimal policy still obeys the displayed inequalities; the proof does not use an exact maximizing selector. Q.E.D.

The exact value function is convex and nondecreasing in d , as a supremum of affine functions with nonnegative slopes. This observation helps interpret the secants but is not an additional assumption needed in the proof. At a kink, distinct optimal policies may have different service. The finite interval covers that set of service responses; it is not a standard error for a unique derivative.

The numerical queries have error allowance $\epsilon = 10^{-7}$, so the implemented endpoints are $\ell_t = \widehat{W}(d + t) - \epsilon$ and $u_t = \widehat{W}(d + t) + \epsilon$. The response tolerance is $\eta = 10^{-8}$ and $h = 0.01$. Thus each untruncated service secant charges $(2\epsilon + \eta)/h = 2.1 \times 10^{-5}$, in addition to economic curvature or policy switching between the queried benefits. The checker reconstructs the values at all three benefits; it does not read a favorable service interval from the generator's summary and call that a proof.

S.15.2. Participation and robust comparison of discrete instruments

PROOF OF THEOREM 2: For a η -optimal response to item j , operating payoff is at least $W_j(d) - \eta \geq \ell_0 - \eta$. Under payment $\bar{q}_j$ its participation payoff is therefore at least

$$\ell_0 - \eta + [G(x_0) - \ell_0 + C(E) + \eta]_+ - C(E) \geq G(x_0). \quad (\text{S.8})$$

Thus the quoted payment is feasible, including when its positive part is zero. Conversely, any participating response under nonnegative payment q has operating payoff no greater than $W_j(d) \leq u_0$. Therefore $q \geq [G(x_0) - u_0 + C(E)]_+ = q_j$.

Proposition 1 and $b \geq 0$ now imply that the selected item's surplus under its executable payment is at least $\underline{\Pi}_j$. Every participating near-optimal response to every item, with any feasible payment, yields surplus at most $\bar{\Pi}_j$. In particular, these upper bounds allow a rival both the smallest potentially feasible payment and the most favorable potentially feasible service; their joint attainability is not required for a valid upper bound. Strict separation against those bounds proves the first assertion. The no-procurement alternative has value zero and is explicitly included.

For the uniform assertion, fix a candidate and a competitor. Their lower-minus-upper difference is affine in (b, κ) because all service and payment interval endpoints are fixed and $D_\kappa(m)$ is affine. Each point of the price rectangle is a convex combination of its vertices; the affine difference is the same convex combination of its vertex values. Positivity at every vertex implies positivity throughout. The selected candidate's lower surplus relative to zero has the same property. There are finitely many competitors, so the argument applies simultaneously to all of them. *Q.E.D.*

The theorem identifies the optimal discrete instrument, not the exact minimum signing payment within that item. Its executable grant includes numerical and response slack. If an application requires strict participation with an additional grant $\delta > 0$, the same proof works whenever δ is smaller than the minimum certified separation. If exact best responses are imposed, they form a subset of the certified response class. No type-reporting or truth-telling constraint is implied by the complete-information menu.

The fee recipient matters. In this environment the fee is paid to an external enforcement sector; no term FH is added to purchaser revenue. A different institutional recipient changes the purchaser's objective and requires a new comparison. Capacity cost is paid in the separate numeraire, while term enforcement cost is paid directly by the purchaser. Payments do not alter managed wealth, so the service bounds obtained before setting the grant remain applicable after the grant is quoted.

S.15.3. *Verification of the procurement theorem*

For each regime, the menu has $2 \cdot 6 \cdot 8 = 96$ items. The generator and independent checker solve the continuation problem for two regimes, six capacities, eight terms and three benefits, or 288 parameter queries. Each query supplies both initial class values. The checker independently optimizes every later-date action backup on every state, verifies the supplied later policy is feasible and value-attaining to its numerical tolerance, verifies the first action and its value, and compares the entire stored continuation-value array, not only the initial scalar.

The first-date action pairs are finite and the risky-share operator is affine between stored knots. Its maximum on a closed permission interval is at a knot or permission endpoint. All permission endpoints used in this numerical theorem are canonical knots for every admissible consumption–adjustment pair; the checker tests this property rather than rounding a requested permission. The positive class is treated by its closure for upper optimization. Feasible strictly positive first actions are retained in the chosen positive contracts and the regional lower witnesses.

The three reconstructed value intervals yield service and payment intervals through the proved formulas. At each of the four (b, κ) vertices and in each regime, the checker

compares the candidate's lower surplus to the upper surplus of every other item and to zero. It also checks that the same candidate wins at all four vertices. The exact generated gaps and input identities are deposited in `independent_validation.json`; outward-displayed bounds in the main tables are computed from that record. This finite arithmetic certificate, the analytic error allowance, and Theorem 2 establish Theorem 3. Center-point reoptimization alone would not establish the rectangle.

Nominal policy moments give a separate accounting check. With the stored selected policy, the generator evaluates (B, A, H) by policy-specific recursions and checks $W = B + dA - FH$. These moments explain the chosen operating allocation but are not substituted for the all-response service enclosure in the decision certificate. The response tolerance in the theorem concerns the economic agent's response class; the theorem does not infer that every numerical maximizing policy is itself 10^{-8} -optimal merely because a floating-point argmax was returned.

S.15.4. Class-specific exposure and its tail representation

PROOF OF PROPOSITION 4: Add and subtract $Q^{\text{adj}}(a_\sigma^0)$ and $Q^0(a_\sigma^0)$ from $W_\sigma^{\text{adj}} - W_\sigma^0$. Since a_σ is an adjusted optimum,

$$\Omega_\sigma = Q^{\text{adj}}(a_\sigma) - Q^{\text{adj}}(a_\sigma^0) + Q^{\text{adj}}(a_\sigma^0) - Q^0(a_\sigma^0) + Q^0(a_\sigma^0) - W_\sigma^0 \quad (\text{S.9})$$

$$= g_\sigma + K_\sigma O_1 + B_\sigma. \quad (\text{S.10})$$

The reward of the fixed zero-drift action is unchanged between the two regimes, so the middle difference is precisely $K_\sigma O_1$. Feasibility of that action for the zero-adjustment class gives $B_\sigma \leq 0$. Subtracting the two class identities proves (14). Decomposing a signed row as $D^+ - D^-$ and using positivity separately on each part proves (15). *Q.E.D.*

For completeness, let ordered state values be $o_1, \dots, o_M$ and let D_j be a signed kernel difference with total mass zero. Define $T_i = \sum_{j=i}^M D_j$. Since $o_j = o_1 + \sum_{i=2}^j (o_i - o_{i-1})$, interchanging finite sums gives

$$\sum_{j=1}^M D_j o_j = o_1 \sum_{j=1}^M D_j + \sum_{i=2}^M (o_i - o_{i-1}) \sum_{j=i}^M D_j = \sum_{i=2}^M T_i (o_i - o_{i-1}). \quad (\text{S.11})$$

This proves the sufficient opportunity-tail condition in the main paper. Without equal masses the first term must be retained. Substochastic live kernels can be augmented by a zero-option cemetery state to obtain equal total mass; the position of that zero option in an asserted monotone ordering must be valid. This transformation does not add a discharge reward, which is already accounted for in the current payoff.

If $\bar{\delta}_i \leq o_i - o_{i-1} \leq \delta_i$, a lower bound is $\sum_i (T_i^+ \bar{\delta}_i - T_i^- \delta_i)$, with an analogous upper bound. Consequently a sufficient class reversal is obtained when a lower bound on the relative local gain, signed future exposure, and financial-replacement difference exceeds $-\Delta^0$. This is a transferable restriction on the mechanism. The application does not assert that primitive wealth ordering alone proves the required ordering of future adjustment values.

At the reported point solutions the zero-drift financial counterpart is also a zero-adjustment optimizer, and both financial-replacement terms reconstruct as zero to numerical tolerance. Their presence in the proposition remains necessary. A different

parameter or permission can change financial controls, so this pointwise observation cannot authorize their removal from the general formula. The signed exposure is localized by wealth bins in the deposited diagnostic record; negative and positive contributions can be much larger in absolute value than their net difference. This is the economic reason to retain signed transport rather than replace it by unrelated worst-case value errors.

S.15.5. Regional lower witnesses and upper recursions

For a fixed feasible policy, write a value at date $n + 1$ in degree $H - 1$ Bernstein form on the common law parameter. Affineness of the one-period reward and kernel in that parameter gives the degree- H coefficient recursion

$$\beta_{n,i}^\pi = \left(1 - \frac{i}{H}\right) \{R_{n,0}^\pi + K_{n,0}^\pi \beta_{n+1,i}^\pi\} + \frac{i}{H} \{R_{n,1}^\pi + K_{n,1}^\pi \beta_{n+1,i-1}^\pi\}, \quad (\text{S.12})$$

with missing endpoint terms omitted. Reward coefficients multiply the degree-elevation weights in the same way as continuation coefficients. A stopping action has zero live-state kernel and current payoff $G - F$. The terminal condition is the stored terminal array. The checker implements this selected-policy recursion directly on the primitive rows, including the first-date action.

For a law subinterval, de Casteljau restriction transforms the global coefficients without changing the polynomial. At any fixed reward corner, nonnegativity and partition of unity of the Bernstein basis imply that the minimum coefficient is a lower bound and the maximum coefficient an upper bound. A policy feasible under the smaller permissions remains feasible at every larger permission in the box. Its reward/fee dependence is affine while that policy remains fixed. Thus the minimum over all corner coefficients is a valid uniform lower bound for that single policy, and the maximum over the policy bank may be taken only *after* this minimum. The proof never chooses a different policy for each coefficient.

For a difference of two optimized values, let L_j^π be the restricted coefficient vector of a feasible policy for target j and let U_k be upper coefficients for target k . Then

$$\max_{\pi} \min_{\text{corners}, i} (L_{j,i}^\pi - U_{k,i}) - 2\epsilon \leq W_j - W_k \leq \min_{\pi} \max_{\text{corners}, i} (U_{j,i} - L_{k,i}^\pi) + 2\epsilon. \quad (\text{S.13})$$

The two minima/maxima across policies need not use the same policy, but each policy must be held fixed within its corner-and-coefficient operation. This proves the reduction used for the adjusted spread, unadjusted spread, and surrender gain. The independent checker first reconstructs the lower and upper coefficients from primitive arrays and then repeats this reduction; it does not accept their roles on trust.

The count-information and signed-compression induction is retained in Supplement S.10. In the new implementation the endpoint optimizations, count interior maximizations, and signed majorant propagation are separately recoded. Each is tested over the full state space and all relevant reward corners. The claimed regional scope is one complete law cell and the displayed permission/reward box; selected-policy checks at a few interior points are not used as a substitute.

For downward-drift elimination, the independently constructed support starts with all feasible first-date transition rows and repeatedly includes every successor of every available later row. The upward-only continuation intervals use adverse benefit/fee

endpoints and endpoint-law mixtures under positive kernels. The signed difference bound in the main paper then dominates every omitted negative-drift action by its feasible zero-drift counterpart. Backward induction from terminal values gives equality of full and upward-only optima on each reachable set. Date- and state-specific frozen actions are checked for unpaired negative drifts. This recreates the valid R11 reasoning on the separately identified R12 target.

S.15.6. *Immutable inputs and the scope of independent checking*

The canonical manifest names every logical input archive, every physical archive part, its byte hash, and every array’s shape, dtype and content hash. The first-date operator and all later operators are loadable without constructor calls. The archive format allows only nonobject numeric arrays. The loader checks array inventory, finiteness, sorted sparse indices, nonnegative transitions, and the stated subprobability mass bound. Witness identities refer to the expected canonical-manifest hash, so changing a manifest along with its arrays does not pass the original witness check.

The explicit new-target command refuses an existing target directory. The generation command similarly refuses an existing witness-output directory. Reproduction reads committed canonical inputs by default. A separately identified target can be generated in another directory, but its result is not silently attributed to the original target. The original R11 expected manifest and all inherited results remain intact. The R12 comparison file records primitive differences without treating small floating-point differences as an operator-transfer proof.

The independent verifier uses the deposited identities as the initial anchor, then applies semantic tests to the mathematical witnesses. Its region reducer rejects the referee’s destroyed upper array even when the original file hash is not consulted. A separate disposable-file test rejects that mutation by archive identity. Other fixtures test infeasible policies, negative transition weights, excess row mass, altered continuation enclosures, stale favorable summaries, and an incorrect canonical identity. These are actual execution tests; no original witness or remote historical file is modified.

Numerical tolerance for comparing two binary64 reconstructions is not itself the economic error bound. The analytic arithmetic audit supplies a per-value bound, and the larger advertised allowance is charged in the economic intervals. The audit explicitly excludes construction and diffusion error. Agreement between two implementations is supporting evidence against coding mistakes, not a theorem that the underlying real-arithmetic construction is identical to another platform’s constructor. Source commit, target identity, witness identity, verification result, and final deposit commit are recorded as different objects.

REFERENCES

- Alegre, Lucas Nunes, Ana L. C. Bazzan, and Bruno C. da Silva (2022): “Optimistic Linear Support and Successor Features as a Basis for Optimal Policy Transfer,” *Proceedings of Machine Learning Research*, 162, 394–413. []
- Amos, Brandon, Lei Xu, and J. Zico Kolter (2017): “Input Convex Neural Networks,” *Proceedings of Machine Learning Research*, 70, 146–155. [6]
- Barles, Guy, and Panagiotis E. Souganidis (1991): “Convergence of Approximation Schemes for Fully Nonlinear Second Order Equations,” *Asymptotic Analysis*, 4, 271–283. [5]
- Barreto, André, Will Dabney, Rémi Munos, Jonathan J. Hunt, Tom Schaul, Hado van Hasselt, and David Silver (2017): “Successor Features for Transfer in Reinforcement Learning,” *Advances in Neural Information Processing Systems*, 30, [17]

- Brown, David B., James E. Smith, and Peng Sun (2010): "Information Relaxations and Duality in Stochastic Dynamic Programs," *Operations Research*, 58, 785–801. []
- Brumm, Johannes, and Simon Scheidegger (2017): "Using Adaptive Sparse Grids to Solve High-Dimensional Dynamic Models," *Econometrica*, 85, 1575–1612. []
- Cai, Wei, Shuixin Fang, and Tao Zhou (2025): "SOC-MartNet: A Martingale Neural Network for the Hamilton–Jacobi–Bellman Equation without Explicit $\inf H$ in Stochastic Optimal Controls," arXiv:2405.03169v4, March 15. []
- Chang, Hyeong Soo (2021): "On-line Policy Iteration with Policy Switching for Markov Decision Processes," arXiv:2112.02177. [17]
- Csáji, Balázs Csanád, and László Monostori (2008): "Value Function Based Reinforcement Learning in Changing Markovian Environments," *Journal of Machine Learning Research*, 9, 1679–1709. []
- Duffie, Darrell, and Larry G. Epstein (1992): "Stochastic Differential Utility," *Econometrica*, 60, 353–394. [10]
- Epstein, Larry G., and Stanley E. Zin (1989): "Substitution, Risk Aversion, and the Temporal Behavior of Consumption and Asset Returns: A Theoretical Framework," *Econometrica*, 57, 937–969. []
- Han, Jiequn, and Weinan E (2016): "Deep Learning Approximation for Stochastic Control Problems," arXiv:1611.07422. []
- Han, Jiequn, Arnulf Jentzen, and Weinan E (2018): "Solving High-Dimensional Partial Differential Equations Using Deep Learning," *Proceedings of the National Academy of Sciences*, 115, 8505–8510. []
- Heck, Linus, Tim Quatmann, Jip Spel, Joost-Pieter Katoen, and Sebastian Junges (2025): "Generalized Parameter Lifting: Finer Abstractions for Parametric Markov Chains," arXiv:2504.05965v2, August 4. []
- Hornik, Kurt, Maxwell Stinchcombe, and Halbert White (1990): "Universal Approximation of an Unknown Mapping and Its Derivatives Using Multilayer Feedforward Networks," *Neural Networks*, 3, 551–560. [5]
- Jacka, Saul D., and Aleksandar Mijatović (2015): "On the Policy Improvement Algorithm in Continuous Time," arXiv:1509.09041. [2]
- Judd, Kenneth L. (1998): *Numerical Methods in Economics*. Cambridge, MA: MIT Press. []
- Kim, Yeongjong, Yeoneung Kim, Minseok Kim, and Namkyeong Cho (2025): "Neural Policy Iteration for Stochastic Optimal Control: A Physics-Informed Approach," arXiv:2508.01718v1, August 3. []
- Kim, Yeongjong, Minseok Kim, Yeoneung Kim, and Namkyeong Cho (2026): "Physics-Informed Policy Iteration for High-Dimensional Hamilton–Jacobi–Bellman Equations: Interior Error Bounds without Boundary Data," arXiv:2508.01718v2, August 9. []
- Lee, Jae Yong, and Yeoneung Kim (2024): "Hamilton–Jacobi Based Policy-Iteration via Deep Operator Learning," arXiv:2406.10920. []
- Merton, Robert C. (1971): "Optimum Consumption and Portfolio Rules in a Continuous-Time Model," *Journal of Economic Theory*, 3, 373–413. []
- Milgrom, Paul, and Ilya Segal (2002): "Envelope Theorems for Arbitrary Choice Sets," *Econometrica*, 70, 583–601.
- Nemecek, Mark, and Ronald Parr (2021): "Policy Caches with Successor Features," *Proceedings of Machine Learning Research*, 139, 8025–8033. []
- Pearlmutter, Barak A. (1994): "Fast Exact Multiplication by the Hessian," *Neural Computation*, 6, 147–160. [17]
- Qi, Qian (2024): "Neural Foundations," working paper, SSRN 5072524. [6]
- Quatmann, Tim, Christian Dehnert, Nils Jansen, Sebastian Junges, and Joost-Pieter Katoen (2016): "Parameter Synthesis for Markov Models: Faster Than Ever," in *Automated Technology for Verification and Analysis*, Lecture Notes in Computer Science, 9938, 50–67. []
- Raissi, Maziar, Paris Perdikaris, and George Em Karniadakis (2019): "Physics-Informed Neural Networks: A Deep Learning Framework for Solving Forward and Inverse Problems Involving Nonlinear Partial Differential Equations," *Journal of Computational Physics*, 378, 686–707. []
- Stokey, Nancy L., Robert E. Lucas, Jr., and Edward C. Prescott (1989): *Recursive Methods in Economic Dynamics*. Cambridge, MA: Harvard University Press. []
- Junges, S., E. Ábrahám, C. Hensel, N. Jansen, J.-P. Katoen, T. Quatmann, and M. Volk (2019): "Parameter Synthesis for Markov Models: Covering the Parameter Space," arXiv:1903.07993, revised November 2023. []